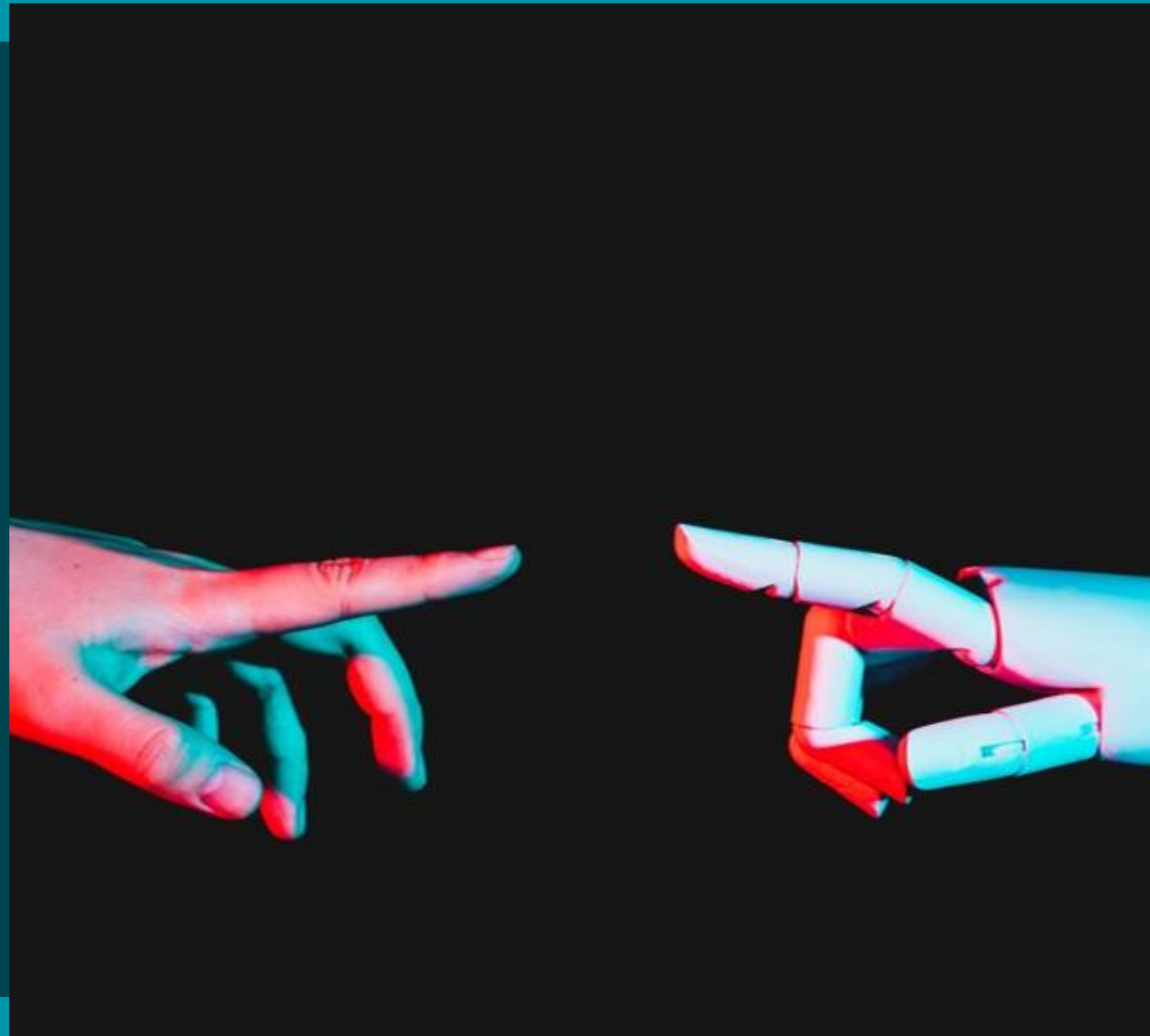


Responsible AI

Introduction to Machine Learning in
Healthcare



Questions from last week?

What is responsible AI ?

Why do we need responsible AI ?

AI's Impact in Society

AI can significantly impact society due to its large range of applications

Some dimensions



AI & Democracy



AI & Future of work



AI & Rights equality



AI & Medicine



AI's Impact in Society

AI can significantly impact society due to its large range of applications

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AI & Future of work



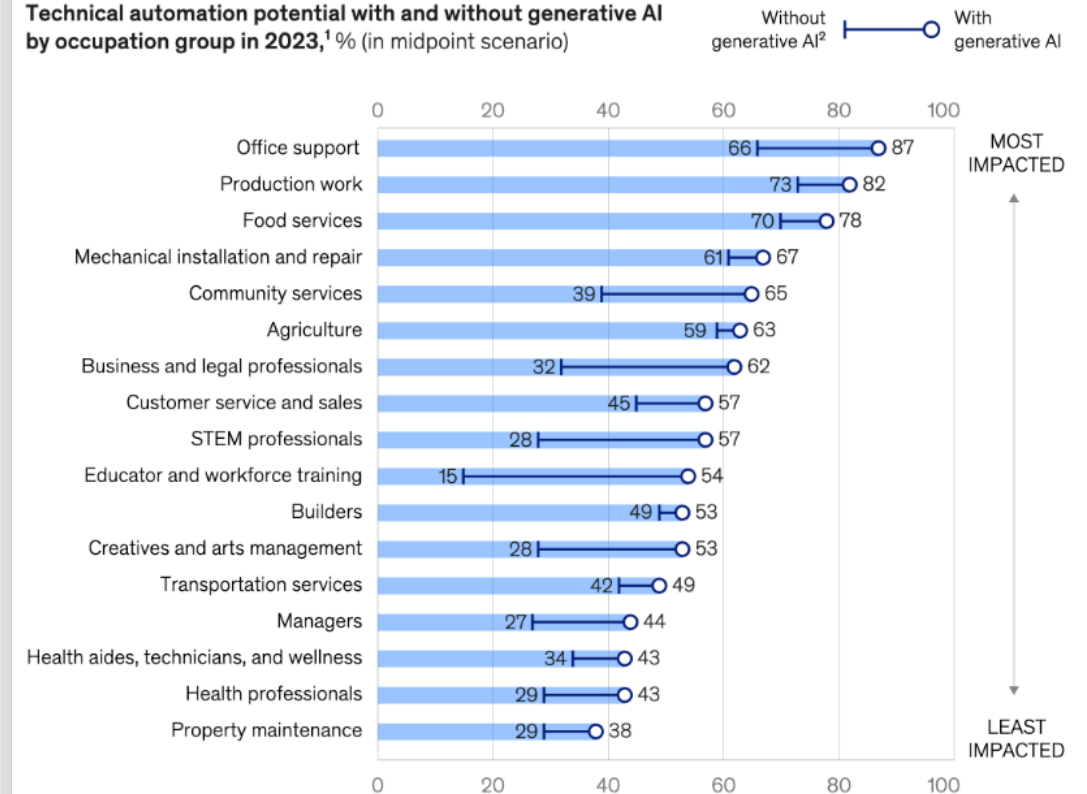
AI & Rights equality



AI & Medicine

Advances in AI's technical capabilities could have the most impact on activities performed by educators, professionals, and creatives.

Technical automation potential with and without generative AI by occupation group in 2023,¹ % (in midpoint scenario)



Note: Figures may not sum to 100%, because of rounding.

¹Overall technical automation potential, comparison in midpoint scenarios.

²Previous assessment of work automation before the rise of generative AI.

Source: McKinsey Global Institute analysis

McKinsey & Company

AI's Impact in Society

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Some dimensions



AI & Democracy



AI & Future of work



AI & Rights equality

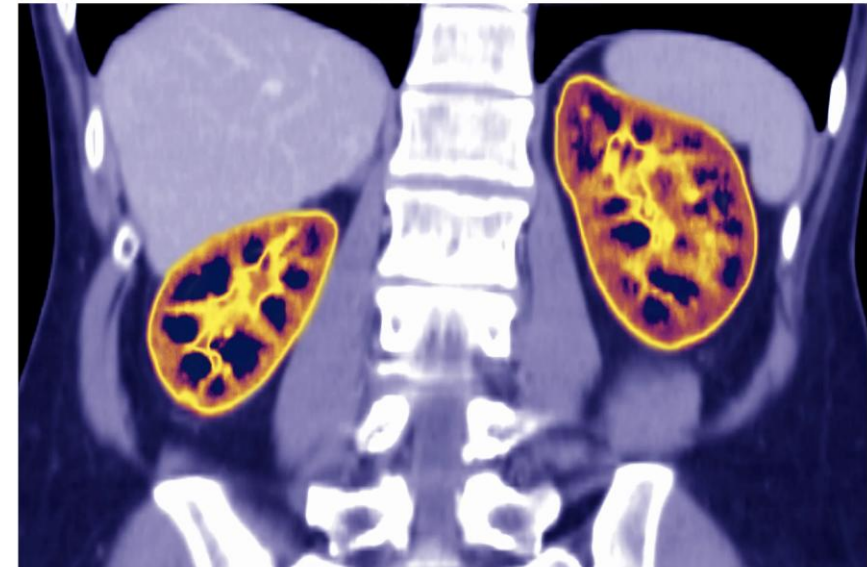


AI & Medicine

TOM SIMONITE BUSINESS OCT 26, 2020 7:00 AM

How an Algorithm Blocked Kidney Transplants to Black Patients

A formula for assessing the gravity of kidney disease is one of many that is adjusted for race. The practice can exacerbate health disparities.



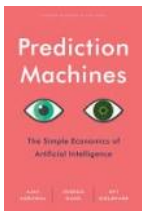
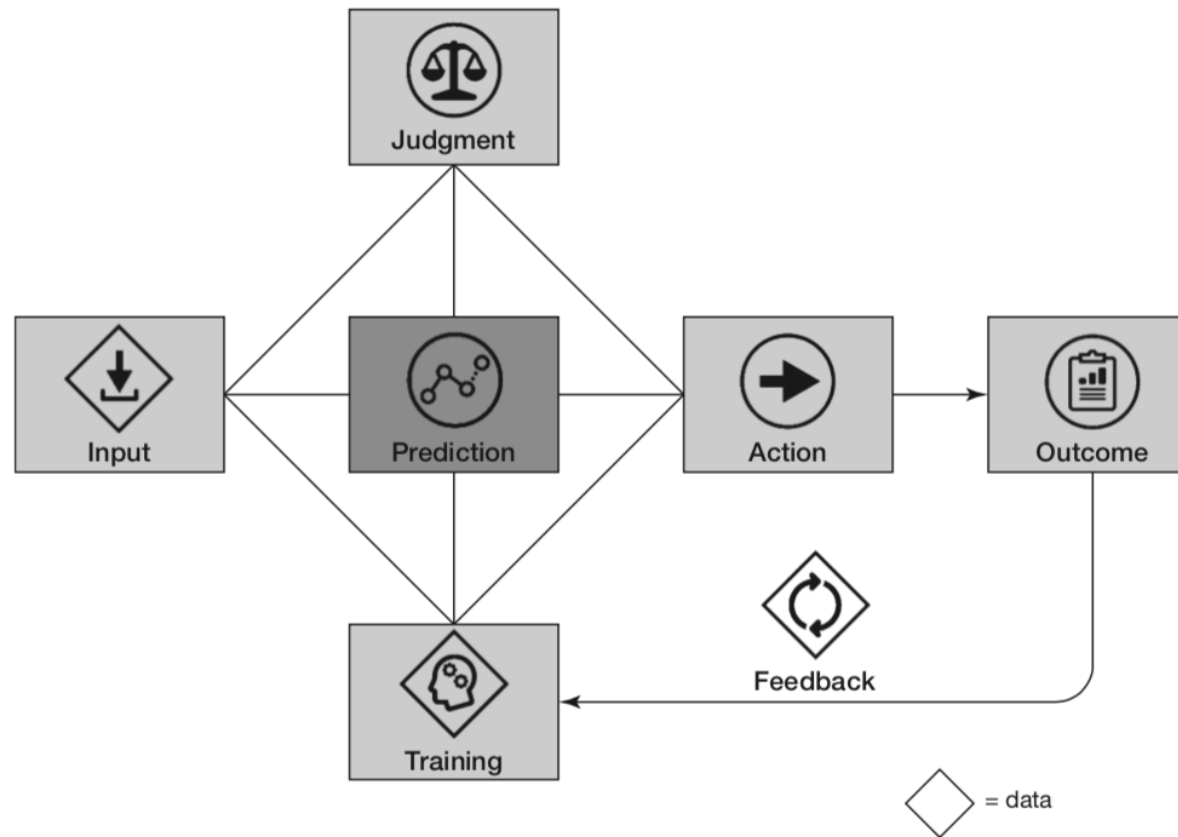
A score known as eGFR aims to reflect the seriousness of a patient's kidney disease. PHOTOGRAPH: JAMES CAVALLINI/SCIENCE SOURCE

 [The AI Database →](#)

APPLICATION: [RECOMMENDATION ALGORITHM](#), [ETHICS](#)

END USER: [BIG COMPANY](#), [SMALL COMPANY](#) SECTOR: [HEALTH CARE](#), [RESEARCH](#)

AI impact's society because we take decisions based on algorithm results... Let's have a look into the anatomy of a decision



With the fast pace of technology and its expected impact in society what can we do?

With the fast pace of technology and its expected impact in society what can we do?

Pause

← All Open Letters

(didn't work)

Pause Giant AI Experiments: An Open Letter

We call on all AI labs to immediately pause for at least 6 months the training of AI systems more powerful than GPT-4.

Signatures

33707

Add your
signature

Published

March 22, 2023

AI systems with human-competitive intelligence can pose profound risks to society and humanity, as shown by extensive research^[1] and acknowledged by top AI labs.^[2] As stated in the widely-endorsed [Asilomar AI Principles](#), *Advanced AI could represent a profound change in the history of life on Earth, and should be planned for and managed with commensurate care and resources*. Unfortunately, this level of planning and management is not happening, even though

<https://futureoflife.org/open-letter/pause-giant-ai-experiments/>

With the fast pace of technology and its expected impact in society what can we do?

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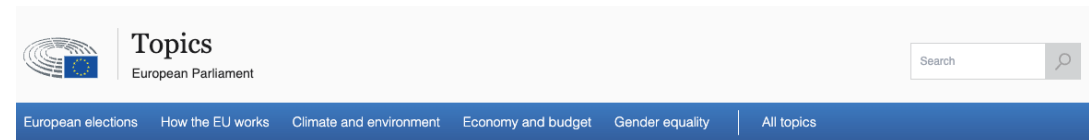
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Regulate



EU AI Act: first regulation on artificial intelligence

The use of artificial intelligence in the EU will be regulated by the AI Act, the world's first comprehensive AI law. Find out how it will protect you.

Published: 08-06-2023

Last updated: 18-06-2024 - 16:29

6 min read

Table of contents

- [AI Act: different rules for different risk levels](#)
- [Transparency requirements](#)
- [Supporting innovation](#)
- [Next steps](#)
- [More on the EU's digital measures](#)

<https://futureoflife.org/open-letter/pause-giant-ai-experiments/>

<https://www.europarl.europa.eu/topics/en/article/20230601STO93804/eu-ai-act-first-regulation-on-artificial-intelligence>

Some highlights regarding the AI Act

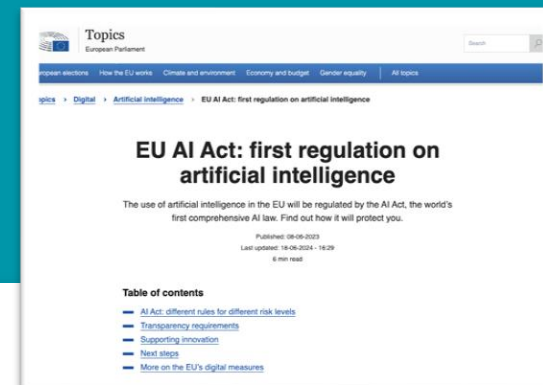
What is AI?

Initially proposed by the EU commission ¹

- “‘artificial intelligence system’ (AI system) means **software** that is developed with one or more of the techniques and approaches listed in Annex I and can, for a given set of **human-defined objectives**, generate outputs such as **content**, **predictions**, **recommendations**, or decisions influencing the environments they interact with”
 - Annex I (Machine learning, Logic and knowledge-based approaches and statistical approaches)

Amendments ²

- “‘artificial intelligence system’ (AI system) means a **machine-based system** that is designed to operate with **varying levels of autonomy** and that can, for **explicit or implicit objectives**, generate outputs such as predictions, recommendations, or decisions, that influence physical or virtual environments;”

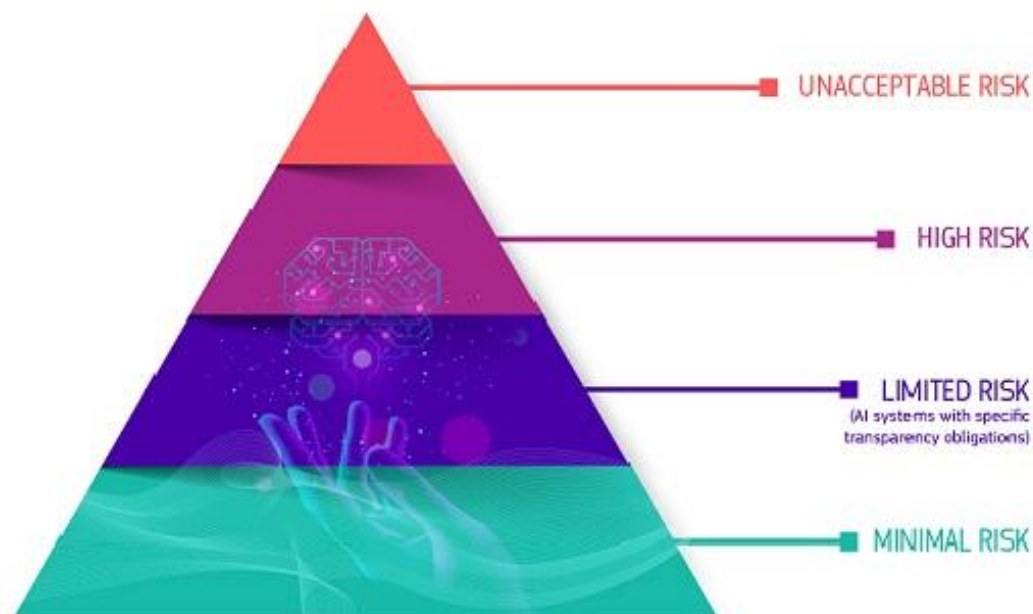


¹EU Proposal - Laying down harmonized rules on artificial intelligence (artificial intelligence act) and amending certain union legislative acts (<https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A52021PC0206>)

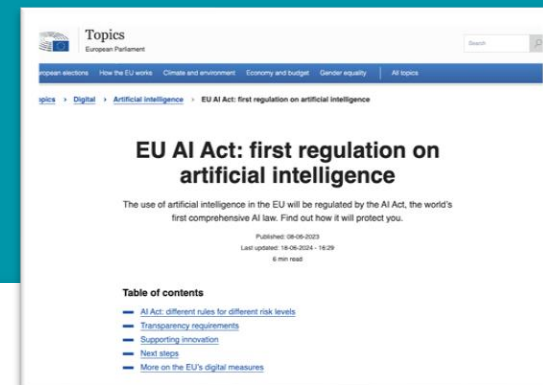
²Amendments adopted by the European Parliament on 14 June 2023 on the proposal for a regulation of the European Parliament and of the Council on laying down harmonised rules on artificial intelligence (Artificial Intelligence Act) and amending certain Union legislative acts (https://www.europarl.europa.eu/doceo/document/TA-9-2023-0236_EN.html)

Some highlights regarding the AI Act

- Risk based approach based on the application and not on the technology itself



- There will be liabilities to companies that don't comply with AI Act



Why do we need responsible AI (in medicine)?

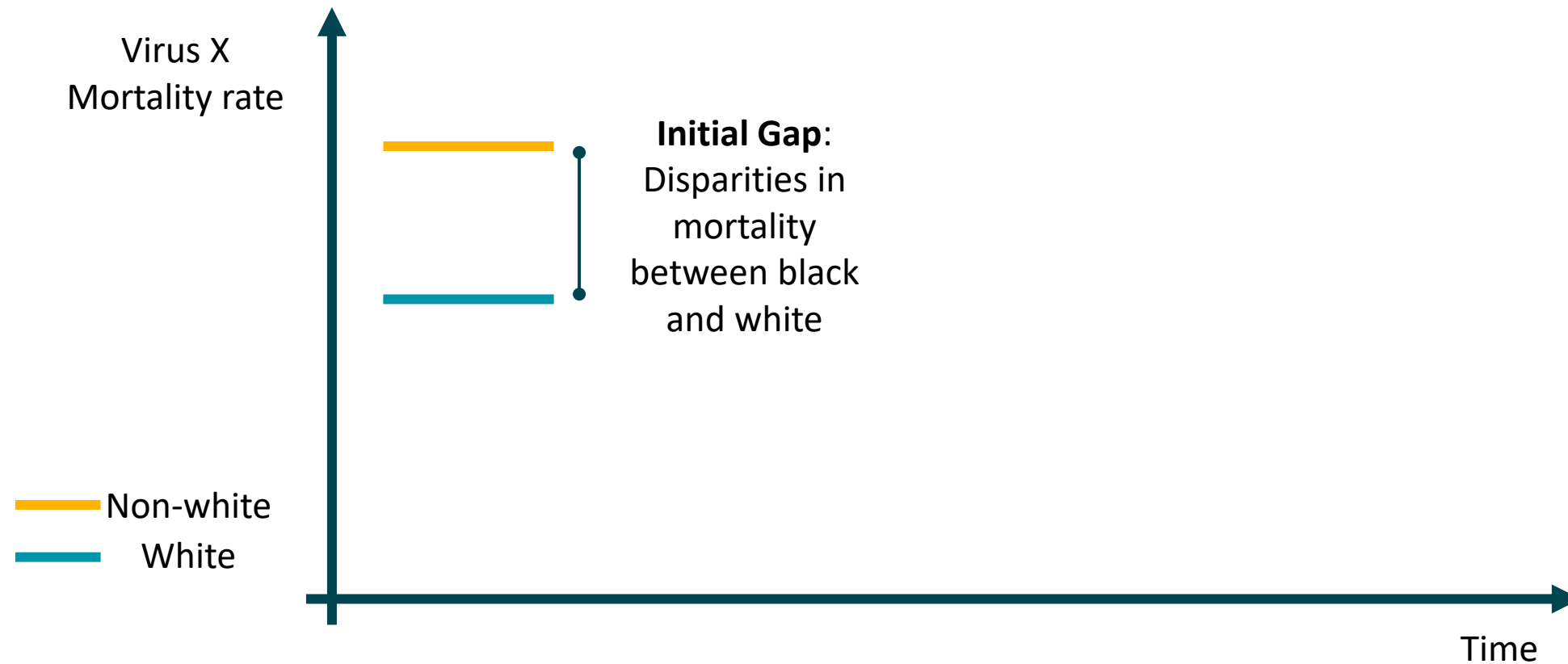
Why do we need responsible AI (in medicine)?

Use Case

- Virus X has high mortality rate on people with certain medical condition
- A new drug is effective, but it is very expensive
- DGS decides to create a ML model using data from EHR and Social Media to predict people at risk to prioritize who must get the treatment

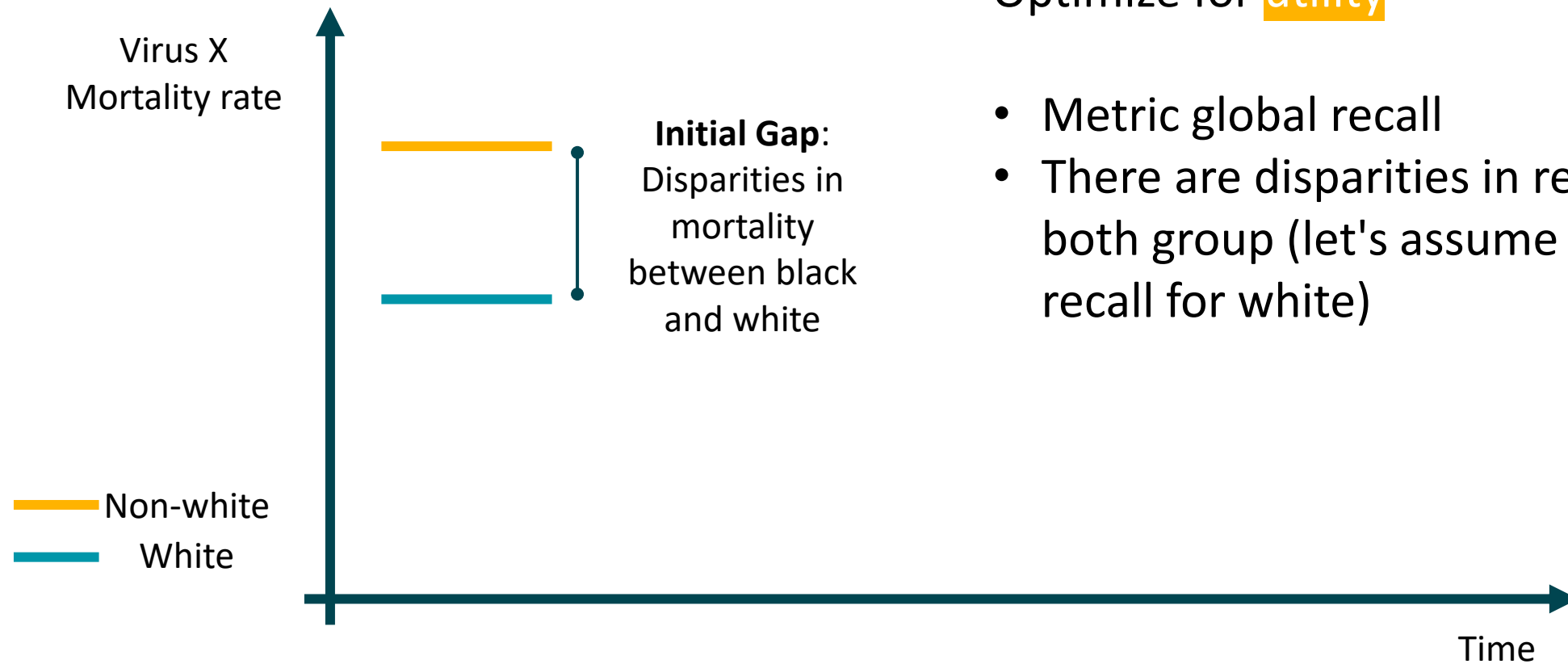
Why do we need responsible AI (in medicine)?

Use Case



Why do we need responsible AI (in medicine)?

Use Case



Optimize for utility

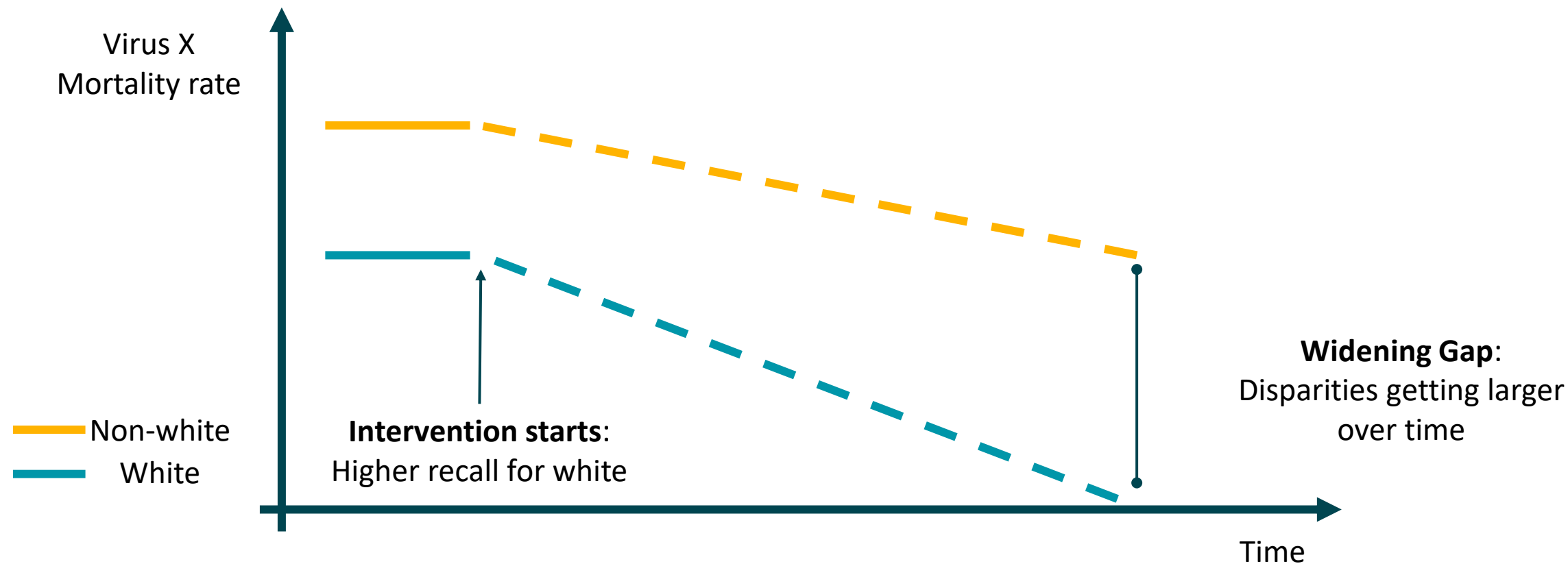
- Metric global recall
- There are disparities in recall for both group (let's assume higher recall for white)

Recall

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

Why do we need responsible AI (in medicine)?

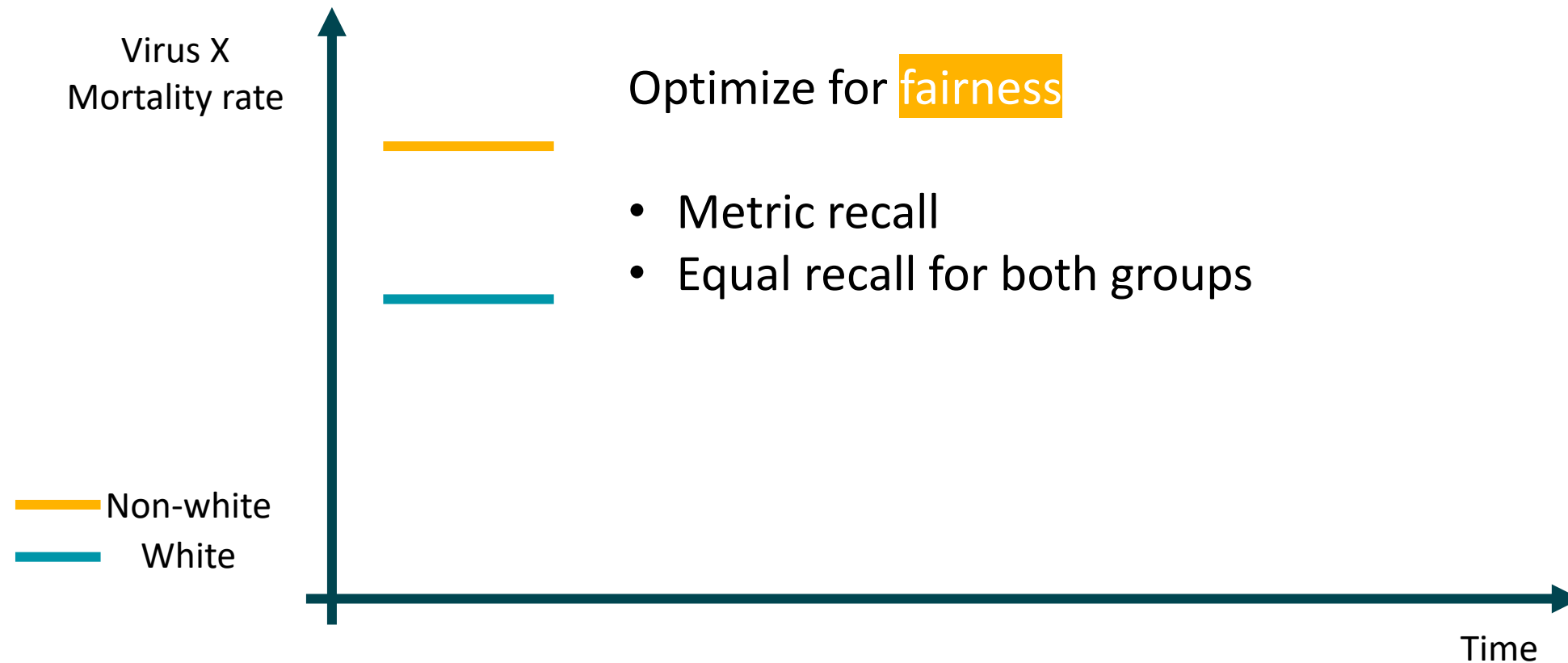
Use Case



Illustrative example assuming intervention is effective

Why do we need responsible AI (in medicine)?

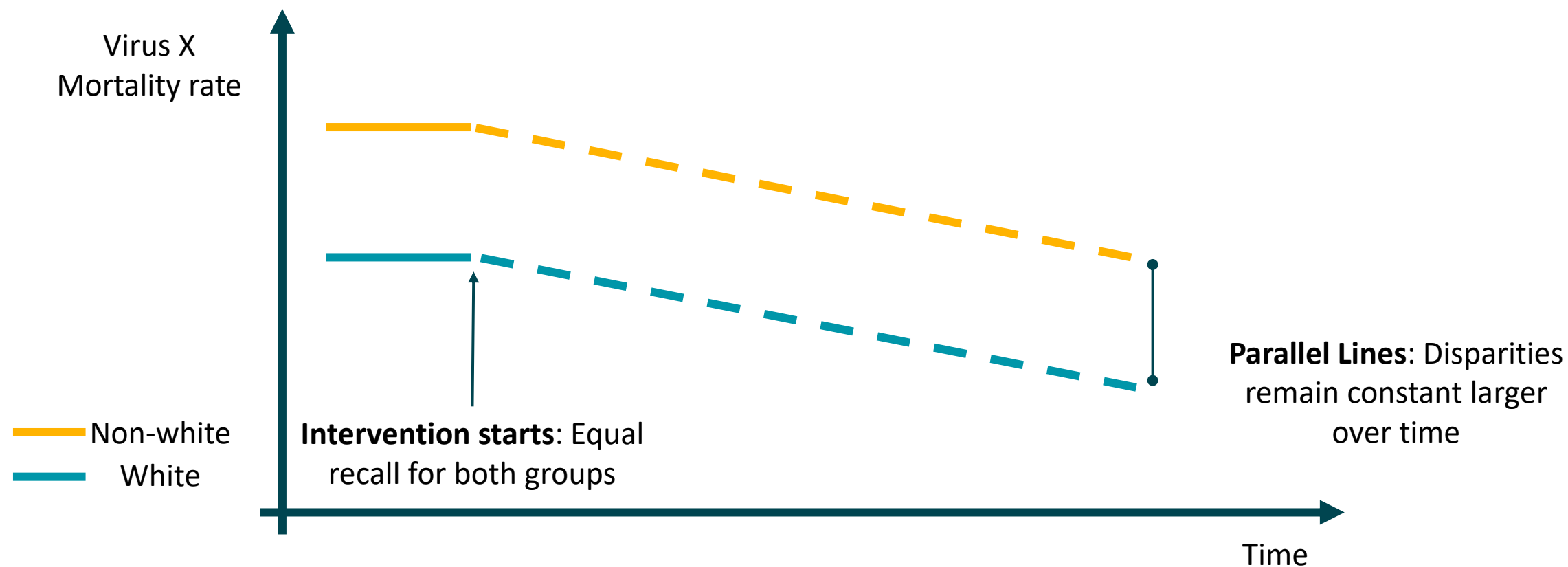
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Illustrative example assuming intervention is effective

Why do we need responsible AI (in medicine)?

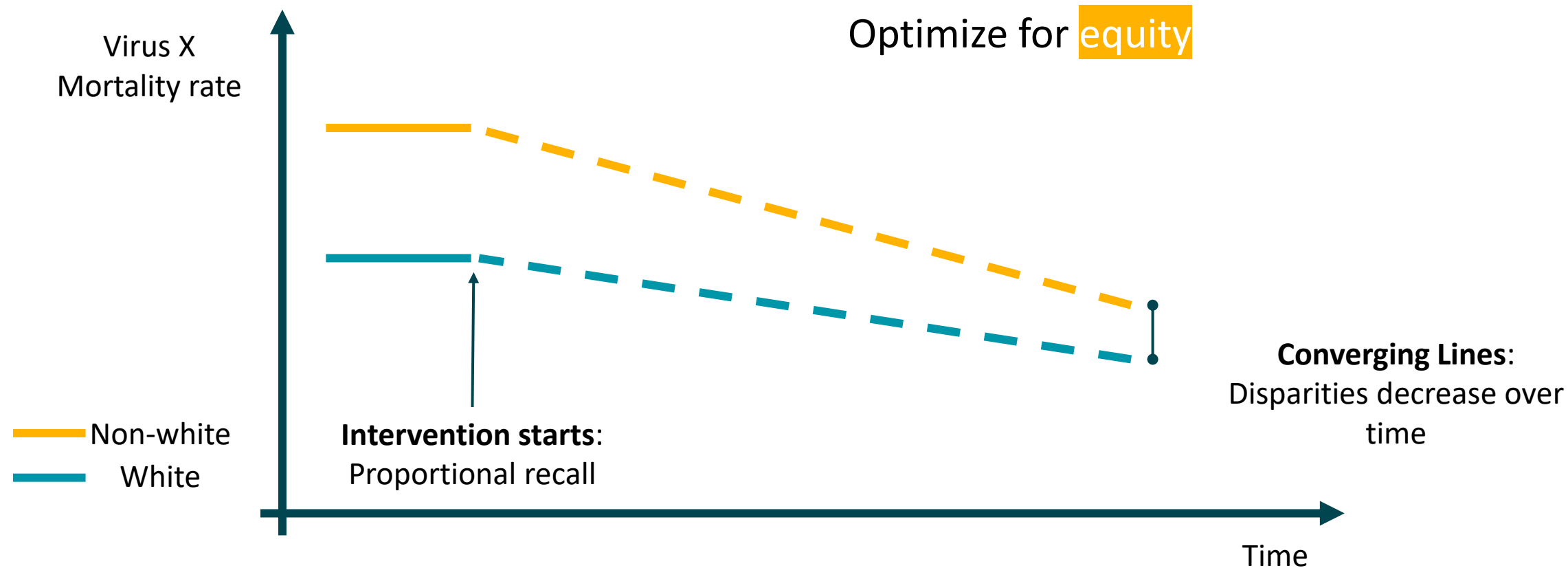
Use Case



Illustrative example assuming intervention is effective

Why do we need responsible AI (in medicine)?

Use Case



Illustrative example assuming intervention is effective

What is responsible AI ?

A structured process to develop and commercialize AI products



- **Responsible AI** is about implementing principles and best practices for building safe, secure, ethical and trustworthy AI systems.

Key concepts in Responsible AI

Every technological company also has its own framework, but the key concepts are the same between them

Accountability

**Transparency and
explainability**

Fairness

Inclusiveness

Privacy and security

Key concepts in Responsible AI

Every technological company also has its own framework, but the key concepts are the same between them

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Transparency & Explainability

This principle is about transparency and responsible disclosure around AI systems to ensure that people understand when they are engaging with them and can challenge outcomes.

Disclose when AI is
being used

Making outputs of
AI interpretable

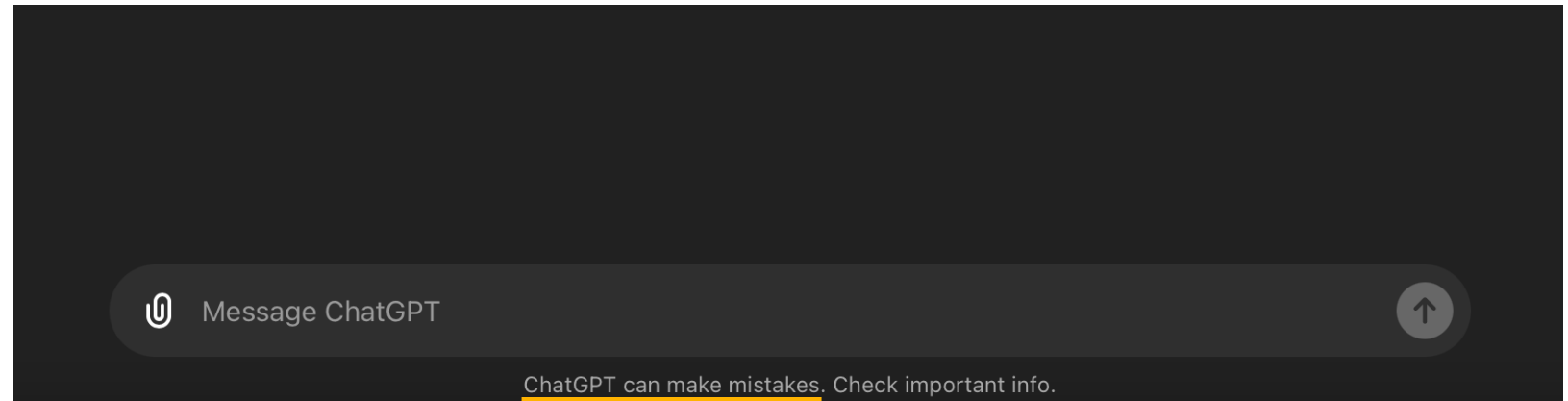
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Disclose when AI is being used

- *People should know when they are interacting with a content generated by AI*
- *Companies should publish summaries of copyrighted data used for training (what data was used in developing the algorithm?)*

Making outputs of AI interpretable



HALLUCINATIONS WITH LLMs

Factuality hallucinations

Sub-Type	User Input	Model Output	Explanation
Factual Inconsistency	Tell me about the first person to land on the Moon.	Yuri Gagarin was the first person to land on the Moon	The LLM's response is factual inconsistency as Neil Armstrong was the first person to land on the Moon , not Yuri Gagarin, who was the first human in space
Factual Fabrication	Tell me about the historical origins of unicorns	Unicorns were documented to have roamed the plains of Atlantis around 10,000 BC, where they were considered sacred creatures and were often associated with royalty.	The LLM's response is a fabricated claim since there's no verified, real-world evidence supporting the existence of unicorns in Atlantis , especially in connection with royalty

Faithfulness hallucinations

Sub-Type	User Input	Model Output	Explanation
Instruction Inconsistency	Translate the following English question into Spanish: "What is the capital of France?"	The capital of France is Paris.	The user explicitly provided an instruction to translate the English question into Spanish. Instead of performing the translation as instructed, the LLM chose to answer the question.
Context Inconsistency	Summarize the following paragraph about the Nile: The Nile is a major north-flowing river in northeastern Africa. It's traditionally considered the longest river in the world. Originating in the Great Lakes region of central Africa , it flows through several countries before emptying into the Mediterranean Sea	The Nile originates from the mountain ranges of central Africa and flows into the Mediterranean.	The LLM's summary incorrectly states that the Nile originates from the "mountain ranges of Central Africa," while the paragraph clearly mentions that the Nile originates in the Great The Lakes region of central Africa.
Logical Inconsistency	Solve the equation $2x + 3 = 11$ step by step.	Step 1: Subtract 3 from both sides to get: $2x = 8$. Step 2: Divide both sides by 2 to get $x = 3$. So, the result of the equation is $x=3$.	The first step correctly isolates the term with x to get $2x = 8$. However, the next step inaccurately divides 8 by 2 to yield a result of $x = 3$, which is inconsistent with the earlier reasoning.

The average rate of hallucinations across major models fell from nearly **38% in 2021 to about 8.2% in 2026.**

CAUSES OF HALLUCINATIONS

Source-Reference Divergence

Exploitation through Jailbreak Prompts

Reliance on Incomplete or Contradictory Datasets

Overfitting and Lack of Novelty

Guesswork from Vague or Insufficiently Detailed Prompts



LLM Hallucinations...

Impossible to eliminate, but possible to reduce

- **Retrieval-augmented generation (RAG)** systems can decrease it by 60–80% by grounding responses in verified documents.
- **Multi-agent verification** catches many errors through redundancy and cross-checking.
- **Strong prompting techniques** significantly improve factual accuracy.

Stop LLM Hallucinations: Reduce Errors by 60–80%

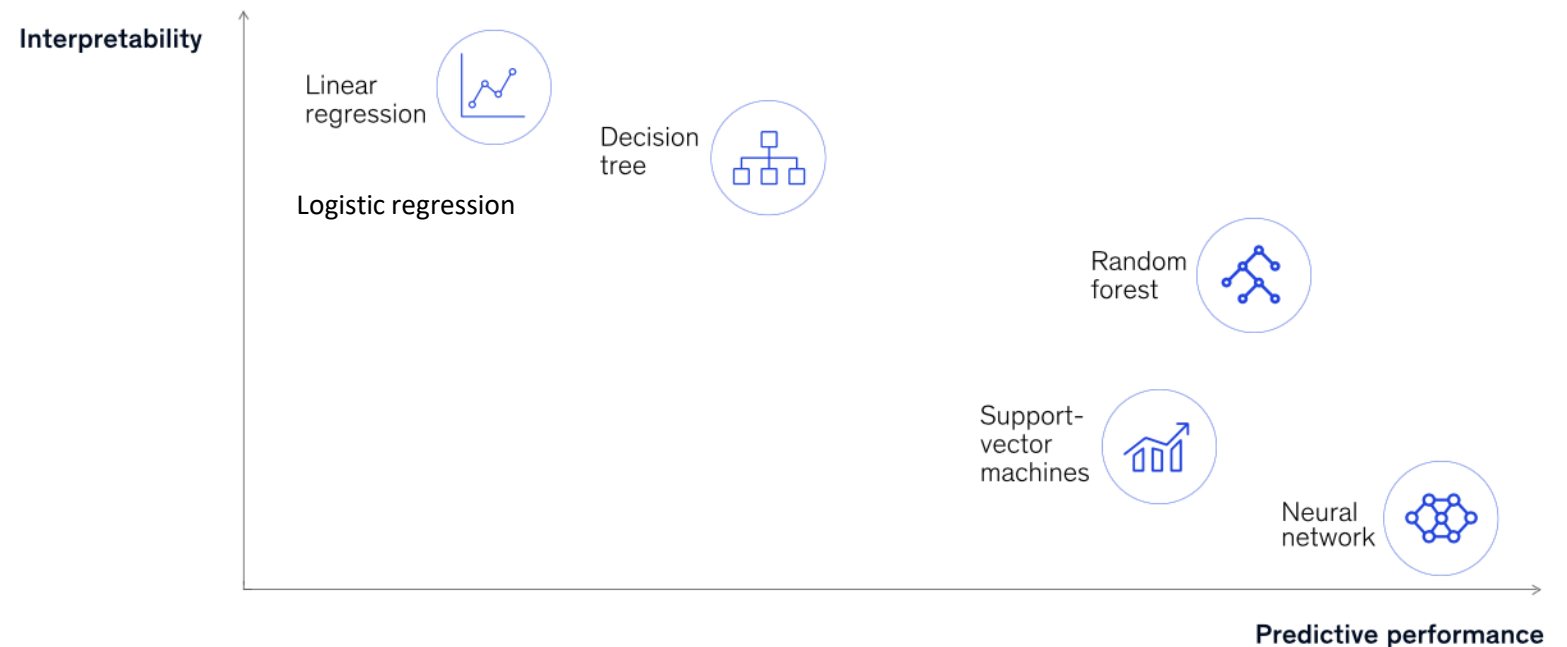
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Disclose when AI is being used

Making outputs of AI interpretable

Models with more predictive power are often more opaque.



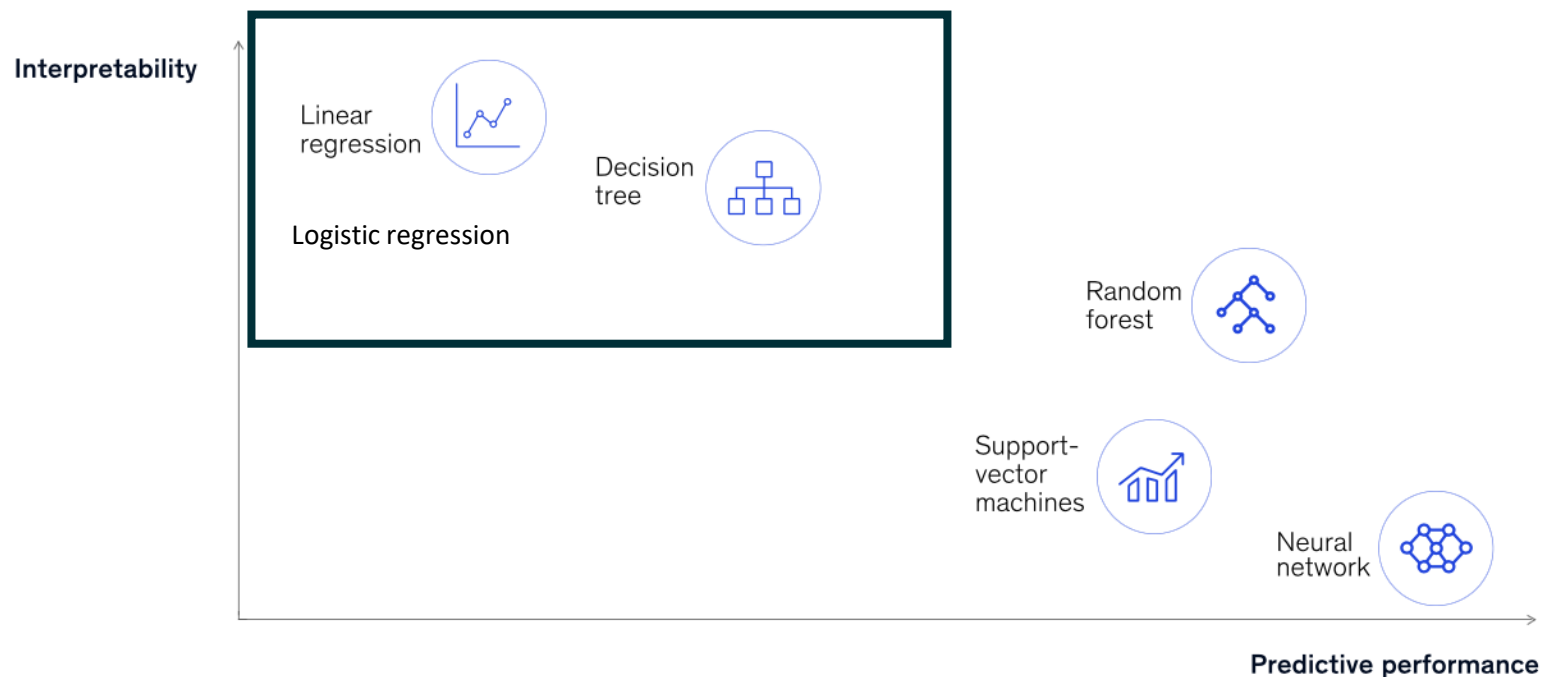
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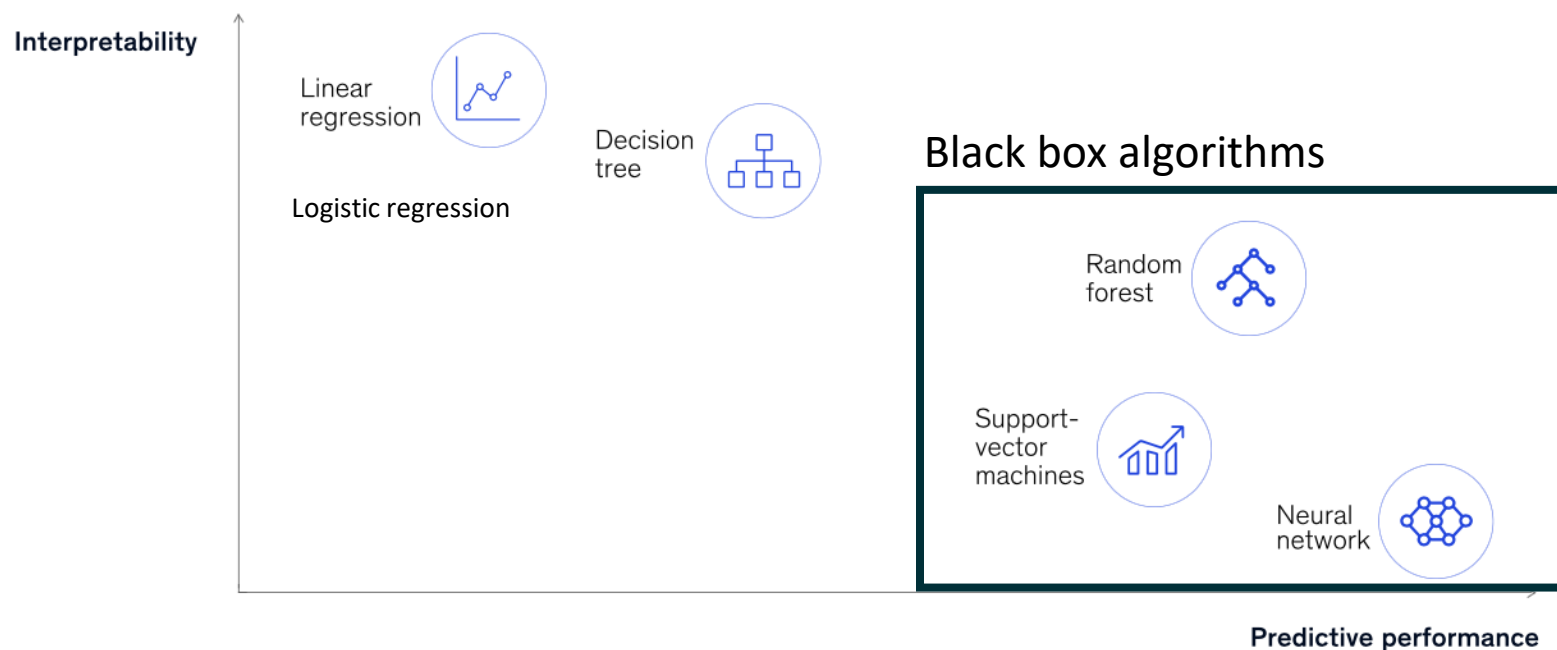
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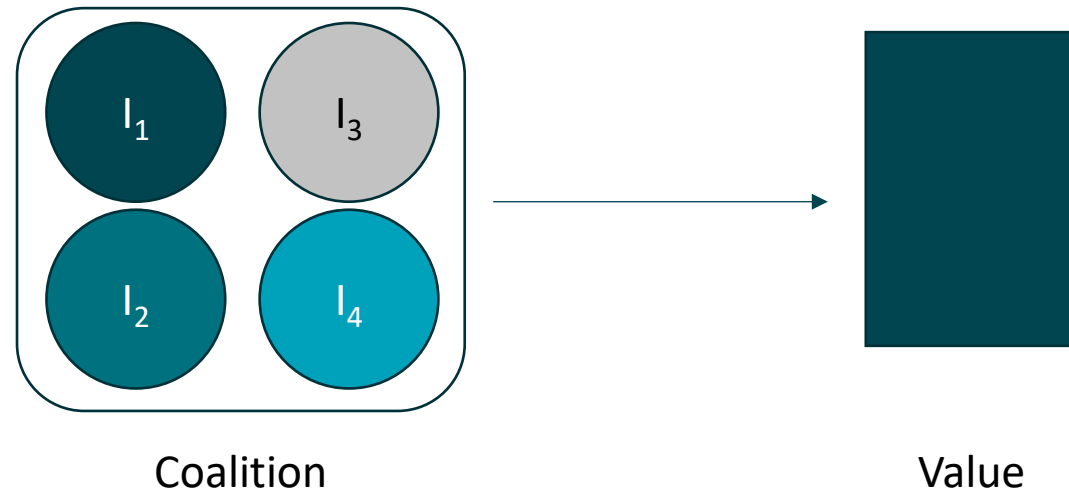
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Disclose when AI is being used

Making outputs of AI interpretable

Shapley values

- Inspired in the game theory



- What is the contribution of everyone to the value?

Transparency & Explainability

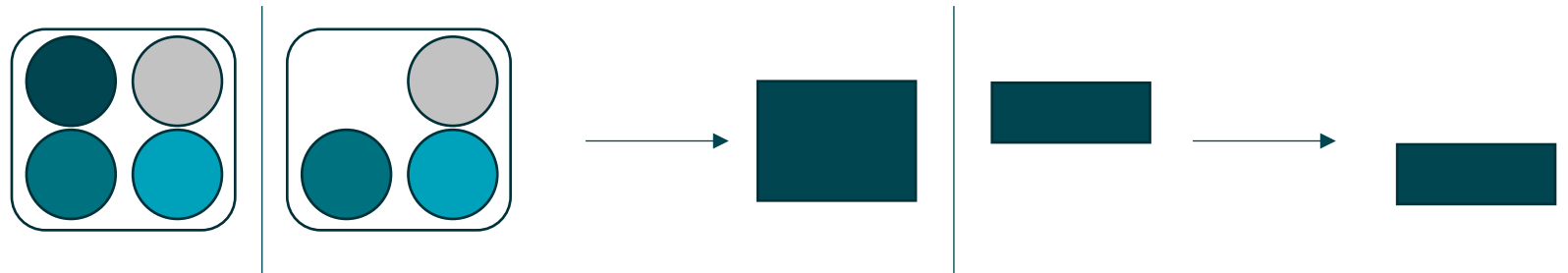
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Disclose when AI is being used

Making outputs of AI interpretable

Shapley values

- Evaluate the value with and without a member of the coalition for all possible combinations



- Calculate the difference between scenarios

Transparency & Explainability

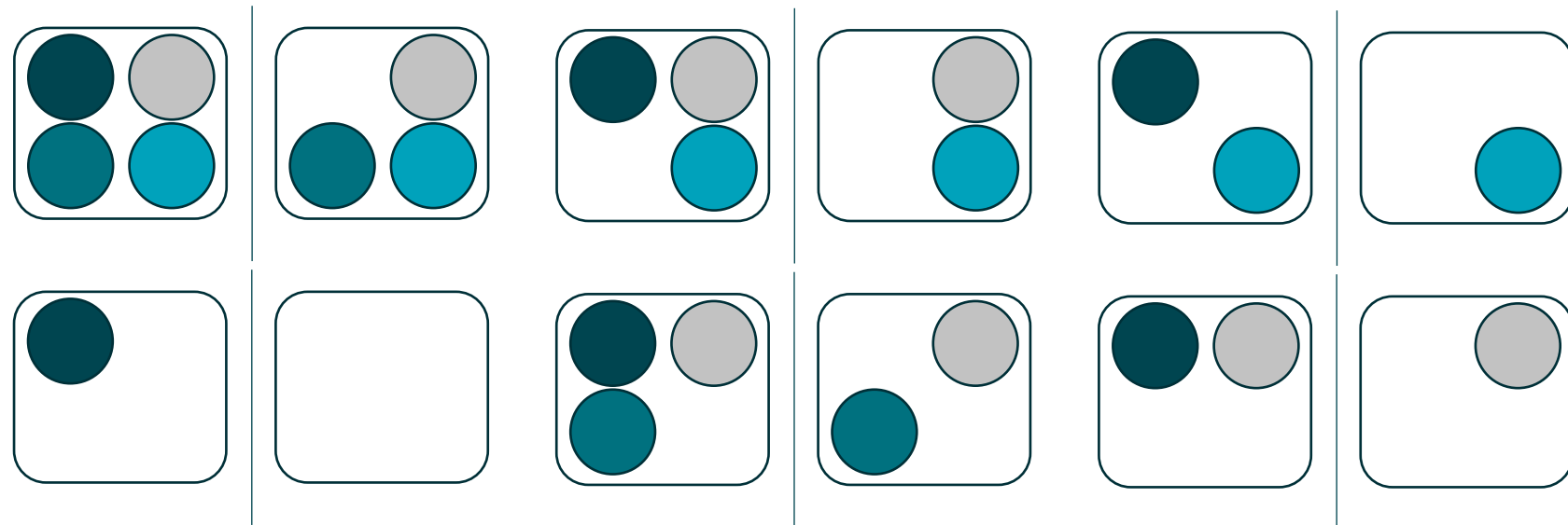
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Making outputs of AI interpretable

Shapley values

- Evaluate the value with and without a member of the coalition for all possible combinations



- The average value of all scenarios is the Shapley value for this member.

Transparency & Explainability

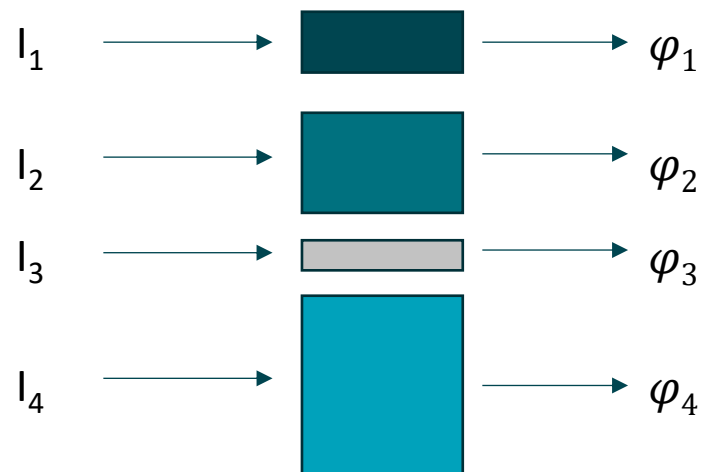
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Disclose when AI is being used

Making outputs of AI interpretable

Shapley values

- Evaluate the value with and without a member of the coalition for all possible combinations



- The average value of all scenarios is the Shapley value for this member.
- The process is repeated for each member

Transparency & Explainability

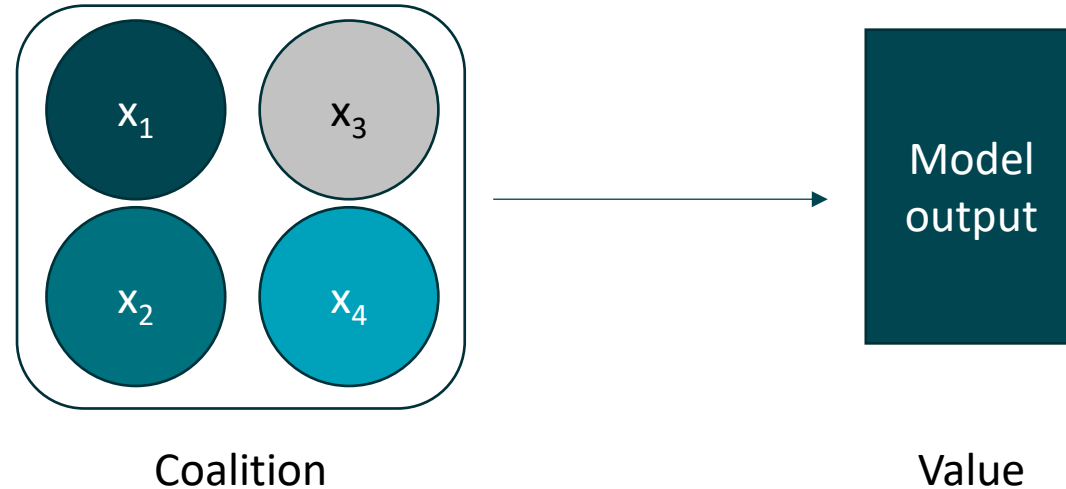
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Disclose when AI is being used

Making outputs of AI interpretable

Shapley values and expandability (SHAP¹)

- The member of the coalition are the variables, and the value is the model output



Transparency & Explainability

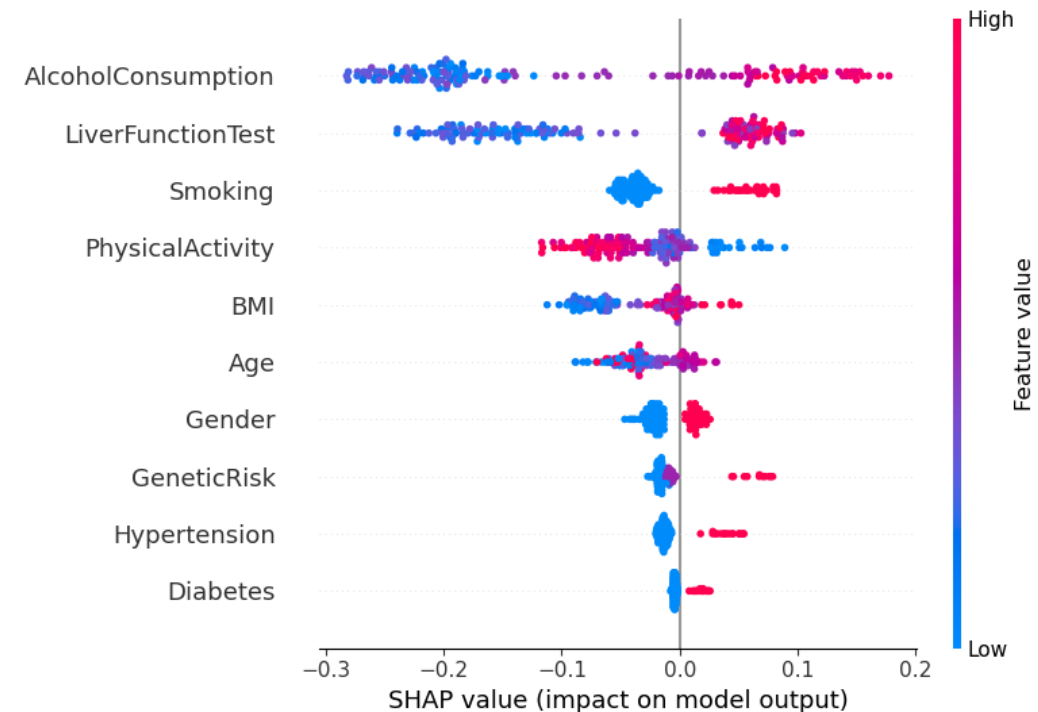
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Disclose when AI is being used

Making outputs of AI interpretable

Shapley values and explainability (SHAP¹)

- The member of the coalition are the variables, and the value is the model output



Detailed video <https://www.youtube.com/watch?v=VB9uV-x0gtg>

¹Original paper <https://doi.org/10.48550/arXiv.1705.07874>

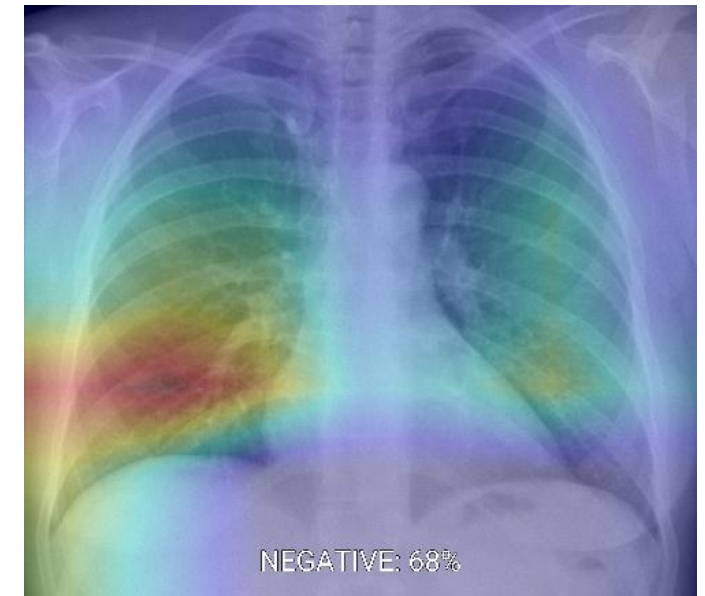
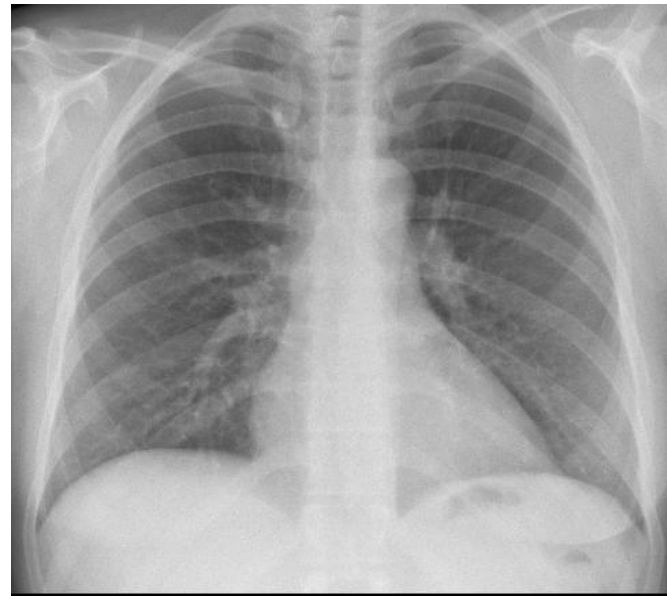
Data: <https://www.kaggle.com/datasets/rabieelkharoua/predict-liver-disease-1700-records-dataset>

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Disclose when AI is being used

Making outputs of AI interpretable





Recap

Let's refresh some concepts

ML model returns above 99% accuracy on real-world data

Is this a good model?

Recap

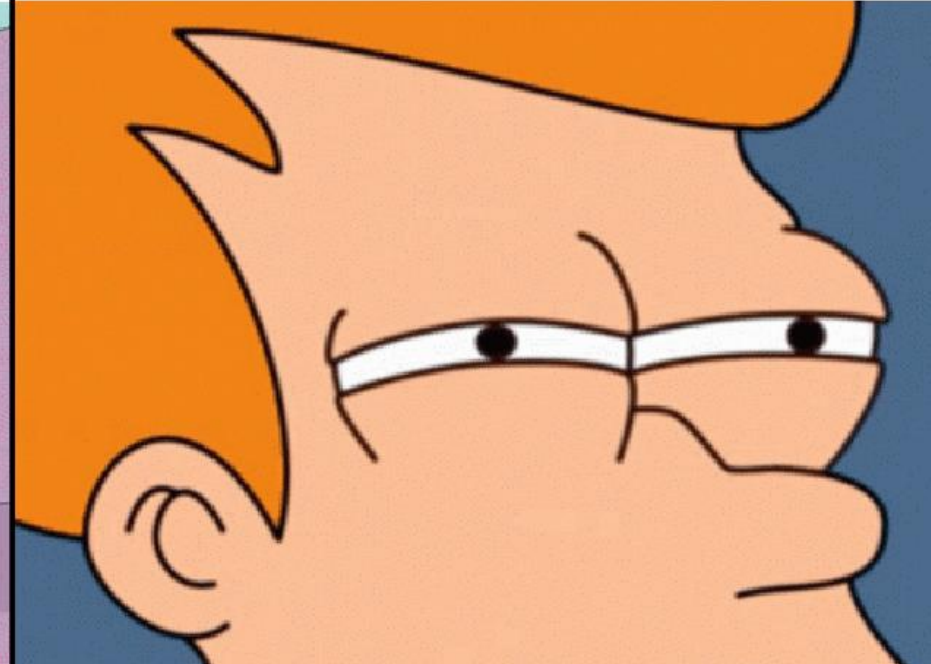
Let's refresh some concepts

ML model returns above 99% accuracy on real-world data

Junior Data Scientist



Senior Data Scientist



Recap

Let's refresh some concepts

Confusion Matrix

		Predicted condition	
		Positive (PP)	Negative (PN)
Actual condition	Total population = P + N		
	Positive (P)	True positive (TP)	False negative (FN)
	Negative (N)	False positive (FP)	True negative (TN)

Calculate metrics



Recap

Let's refresh some concepts

Confusion Matrix

		Predicted condition	
Total population = P + N		Positive (PP)	Negative (PN)
Actual condition	Positive (P)	True positive (TP)	False negative (FN)
	Negative (N)	False positive (FP)	True negative (TN)

Calculate metrics



Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Error Rate

$$\text{Error Rate} = \frac{FP + FN}{TP + TN + FP + FN}$$

Precision

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall

$$\text{Recall} = \frac{TP}{TP + FN}$$

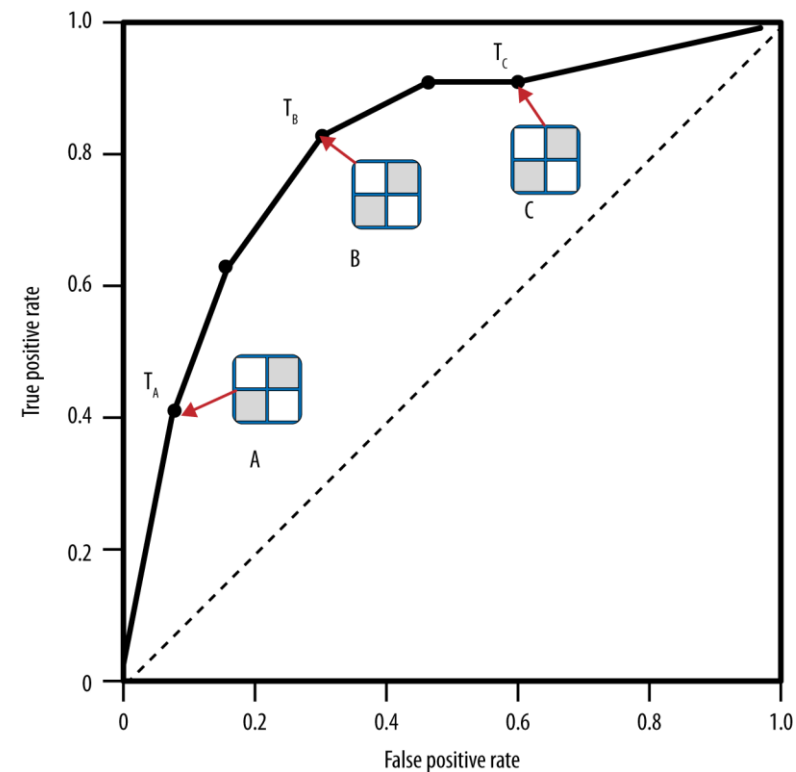
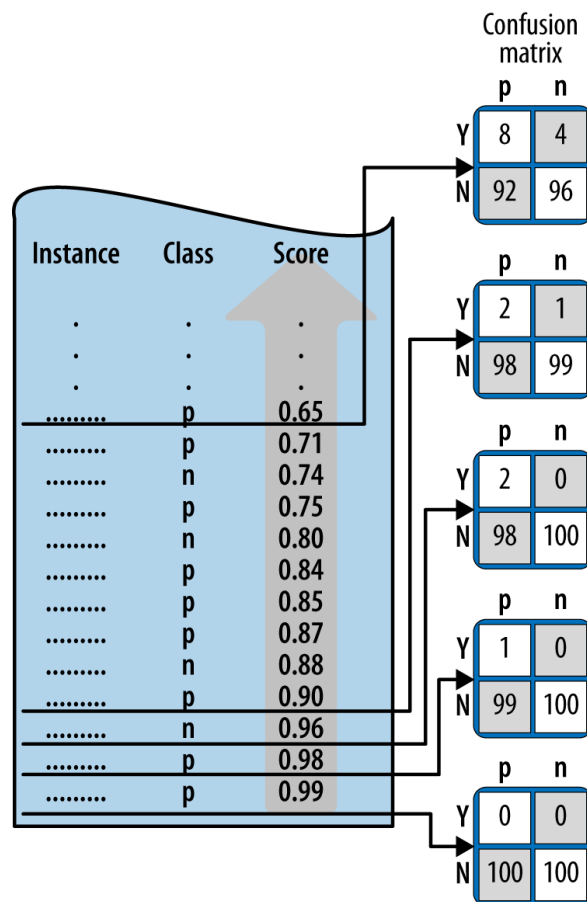
F1 - score

$$\text{F1 - score} = 2 \times \frac{\text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}}$$

Recap

Let's refresh some concepts

ROC Curve



Fairness

Receiver operating curve

- Predict liver disease diagnostics

Index	Score	Label_value
1	0.999800	1
2	0.986034	1
3	0.965328	1
4	0.927179	1
5	0.889940	0
6	0.837430	1
7	0.773571	1
8	0.690866	1
9	0.590255	1
10	0.468367	0
11	0.357743	1
12	0.276027	0
13	0.193268	0
14	0.145628	1
15	0.103324	0
16	0.062255	0
17	0.028513	0

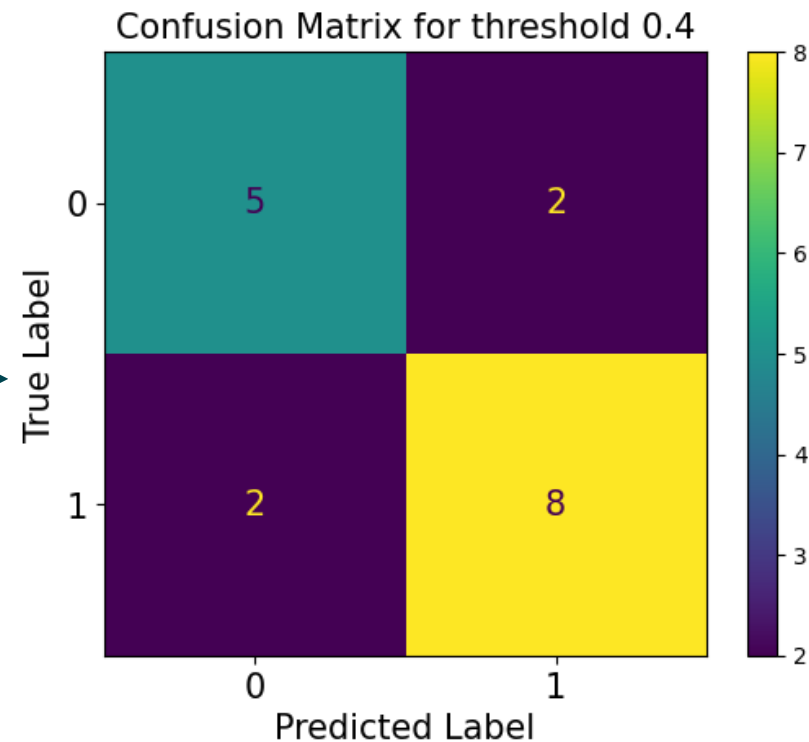
Label: 1 positive, 0 negative

Fairness

Receiver operating curve

- Predict liver disease diagnostics

Index	Score	Label_value
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13	0.193268	0
14	0.145628	1
15	0.103324	0
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$$\text{TPR} = \text{tp} / (\text{tp} + \text{fn}) = 8/10 = 0.8$$
$$\text{FPR} = \text{fp} / (\text{fp} + \text{tn}) = 2/7 = 0.3$$
$$\text{Accuracy} = 13/17 = 0.8$$

Label: 1 positive, 0 negative

Fairness

Receiver operating curve

- Predict liver disease diagnostics

Index	Score	Label_value	Race
1	0.999800	1	White
2	0.986034	1	Non-White
3	0.965328	1	Non-White
4	0.927179	1	White
5	0.889940	0	White
6	0.837430	1	White
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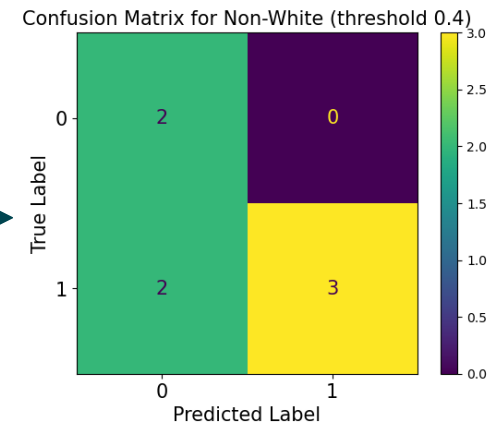
Label: 1 positive, 0 negative

Fairness

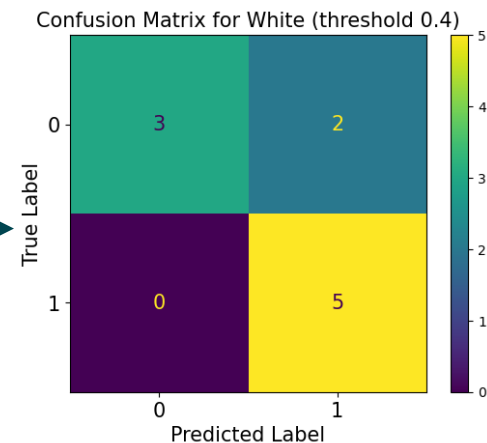
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16	0.062255	0	Non-White
17	0.028513	0	White



$$\text{TPR} = \text{tp} / (\text{tp} + \text{fn}) = 3 / (3 + 2) = 0.6$$
$$\text{FPR} = \text{fp} / (\text{fp} + \text{tn}) = 0 / (0 + 2) = 0$$
$$\text{Accuracy} = 5 / 7 = 0.7$$



$$\text{TPR} = \text{tp} / (\text{tp} + \text{fn}) = 5 / (5 + 0) = 1$$
$$\text{FPR} = \text{fp} / (\text{fp} + \text{tn}) = 2 / (2 + 3) = 0.4$$
$$\text{Accuracy} = 8 / 10 = 0.8$$

Label: 1 positive, 0 negative

Bias

- Bias is about disparate errors against specific sub-groups.

Use Case

Dissecting racial bias in an algorithm used to manage the health of populations

RESEARCH ARTICLE

ECONOMICS

Dissecting racial bias in an algorithm used to manage the health of populations

Ziad Obermeyer^{1,2*}, Brian Powers³, Christine Vogeli⁴, Sendhil Mullainathan^{5*†}

Health systems rely on commercial prediction algorithms to identify and help patients with complex health needs. We show that a widely used algorithm, typical of this industry-wide approach and affecting millions of patients, exhibits significant racial bias: At a given risk score, Black patients are considerably sicker than White patients, as evidenced by signs of uncontrolled illnesses. Remedying this disparity would increase the percentage of Black patients receiving additional help from 17.7 to 46.5%. The bias arises because the algorithm predicts health care costs rather than illness, but unequal access to care means that we spend less money caring for Black patients than for White patients. Thus, despite health care cost appearing to be an effective proxy for health by some measures of predictive accuracy, large racial biases arise. We suggest that the choice of convenient, seemingly effective proxies for ground truth can be an important source of algorithmic bias in many contexts.

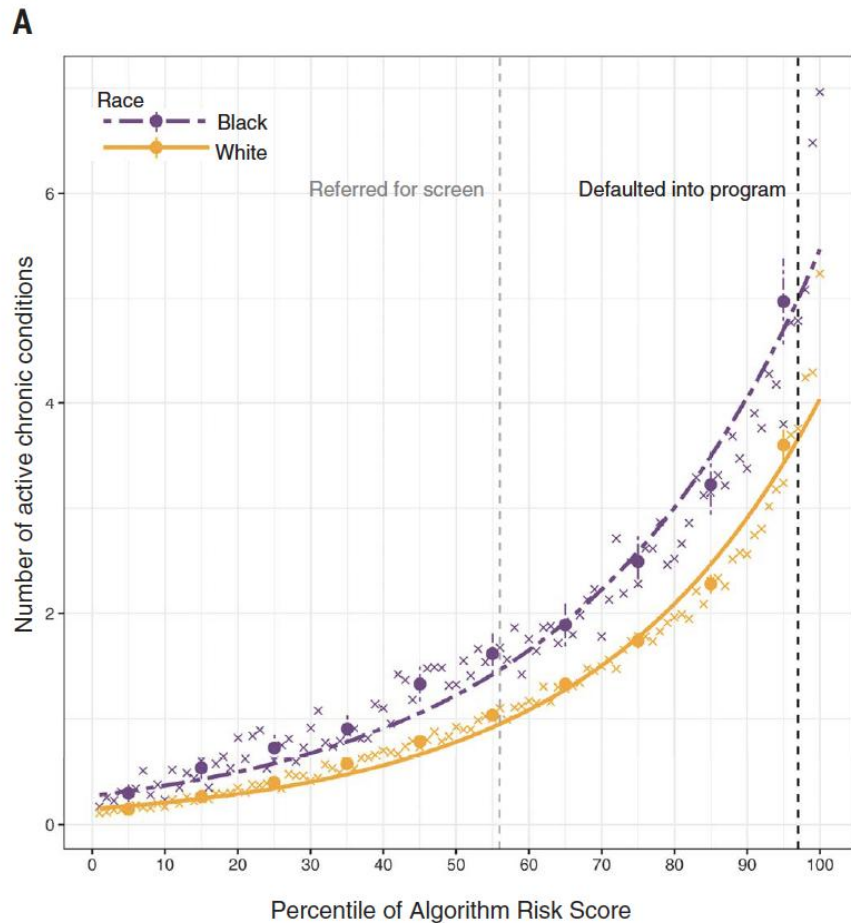
Use Case

Dissecting racial bias in an algorithm used to manage the health of populations

- **Health System Data:** The study uses data from a commercial risk-prediction algorithm deployed in U.S. health systems to manage high-risk patients. The prediction is used to identify patients who should be enrolled in high-risk care management programs.
- **Study Sample:** It includes data from 43,539 White and 6,079 Black primary care patients enrolled in risk-based contracts from 2013 to 2015.
- **Algorithm Data:** The dataset contains the algorithm's predictions and the underlying data, including demographics, diagnoses, medications, and costs.
- **Outcome Measures:** The target variable in the article is future health care costs. Patients above the 97th percentile are automatically identified for enrollment in the program. Those above the 55th percentile are referred to their primary care physician and evaluates the impact of the program.
- **How does racial bias manifest in a widely used commercial prediction algorithm?**

Use Case

Dissecting racial bias in an algorithm used to manage the health of populations

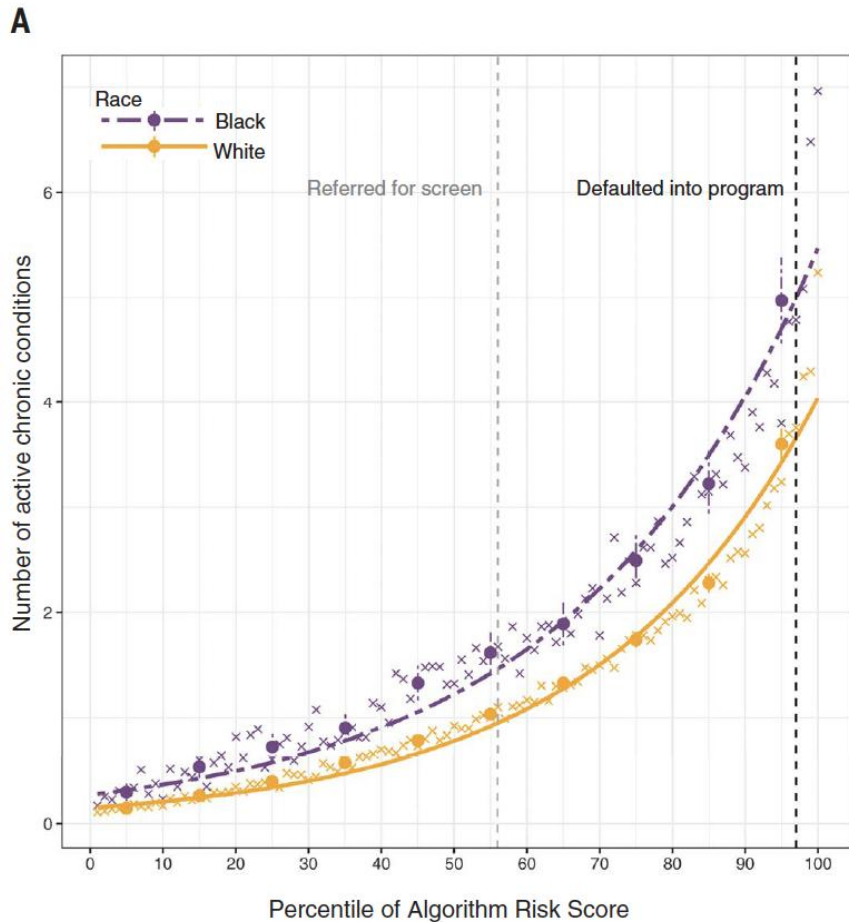


Ziad Obermeyer *et al.* Dissecting racial bias in an algorithm used to manage the health of populations. *Science* **366**,447-453(2019).

DOI:[10.1126/science.aax2342](https://doi.org/10.1126/science.aax2342)

Use Case

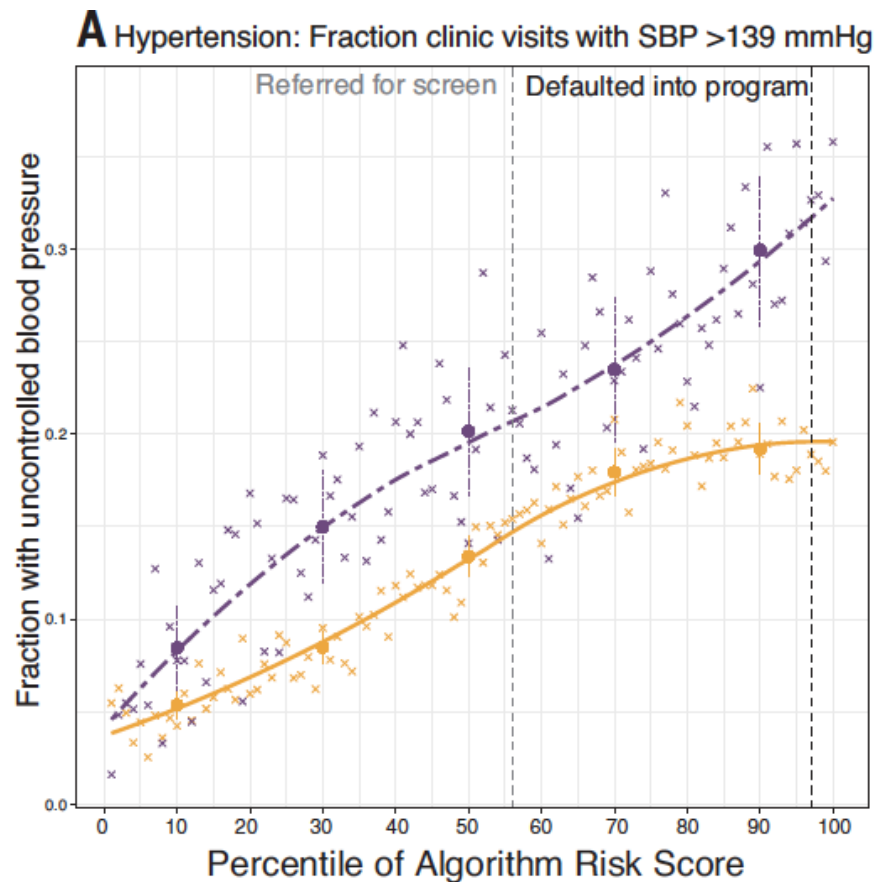
Dissecting racial bias in an algorithm used to manage the health of populations



- At the same level of algorithm-predicted risk, Black patients have a significantly higher burden of chronic illnesses compared to White patients.
- This discrepancy indicates that the algorithm's risk scores do not accurately reflect the actual health status of Black patients, leading to underestimation of their health needs.

Use Case

Dissecting racial bias in an algorithm used to manage the health of populations

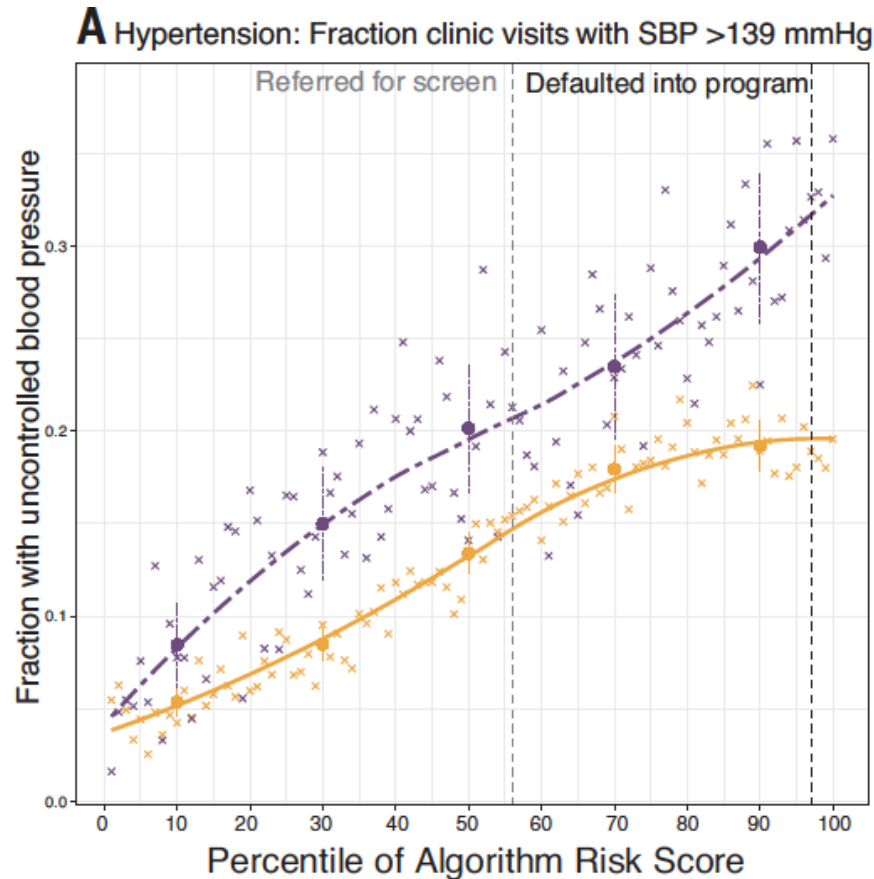


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Use Case

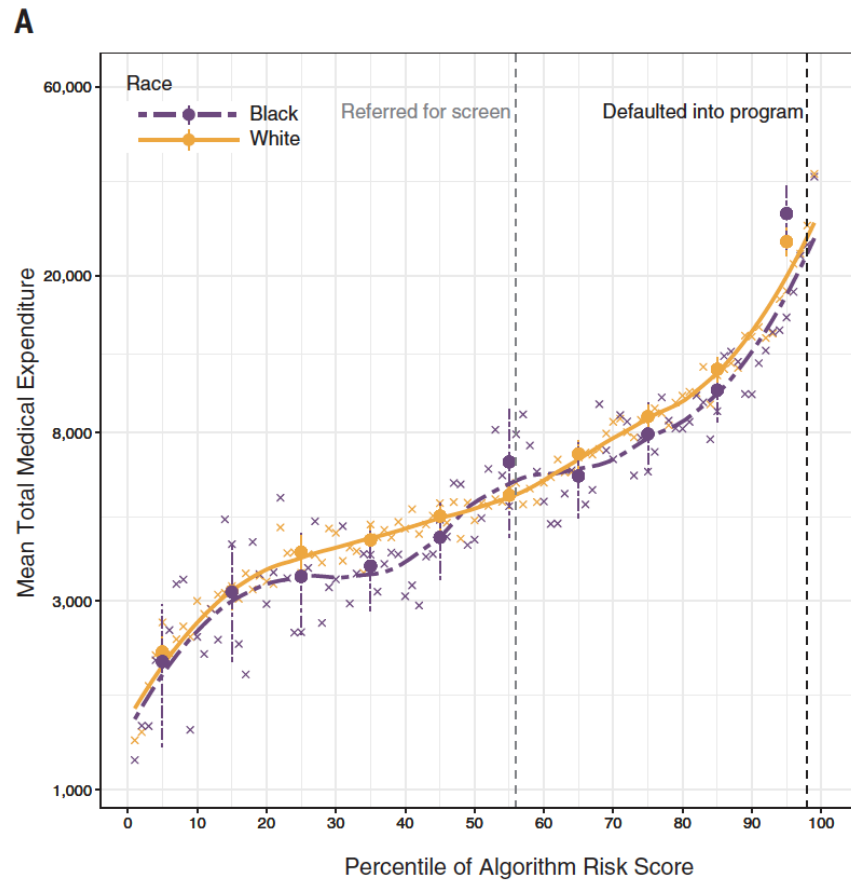
Dissecting racial bias in an algorithm used to manage the health of populations



- This figure shows that Black patients have higher rates of uncontrolled hypertension (systolic blood pressure > 139 mmHg) than White patients at the same algorithm-predicted risk levels.
- This indicates that, despite similar risk scores, Black patients are experiencing more severe hypertension, suggesting that the algorithm underestimates the true health needs of Black patients.

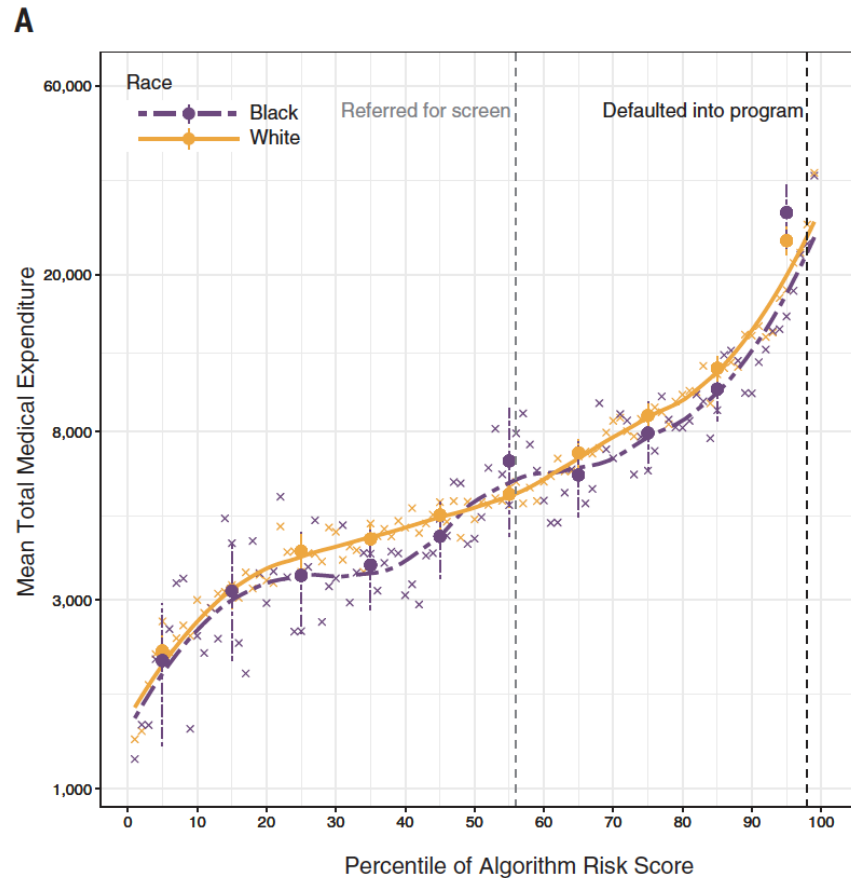
Use Case

Dissecting racial bias in an algorithm used to manage the health of populations



Use Case

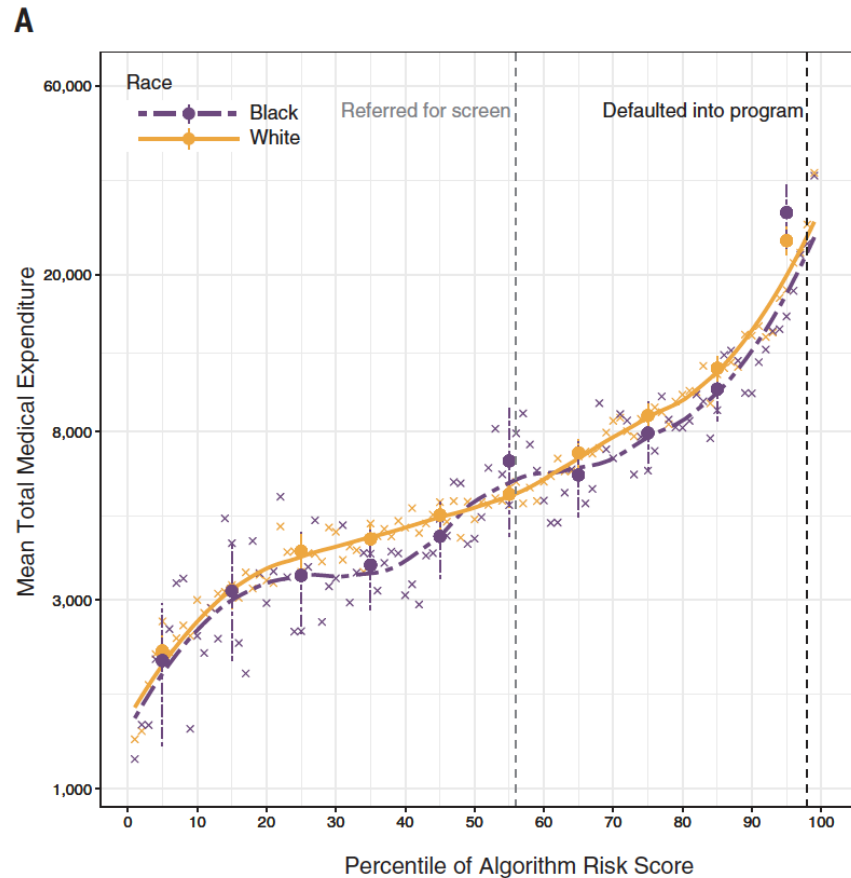
Dissecting racial bias in an algorithm used to manage the health of populations



- Fig A shows that, at each level of predicted risk, Black and White patients have similar total medical expenditures.
- This suggests that, in terms of cost predictions, the algorithm appears well-calibrated across races.

Use Case

Dissecting racial bias in an algorithm used to manage the health of populations



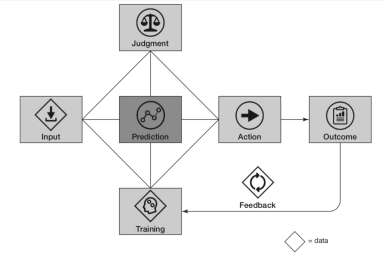
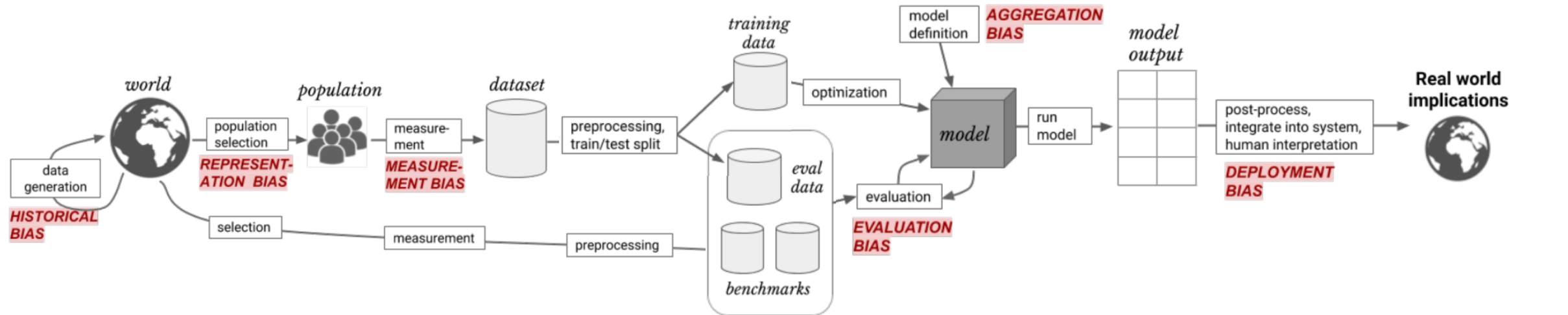
- Fig A shows that, at each level of predicted risk, Black and White patients have similar total medical expenditures.
- This suggests that, in terms of cost predictions, the algorithm appears well-calibrated across races.

Where is bias coming from?

Fairness

Bias in decision making

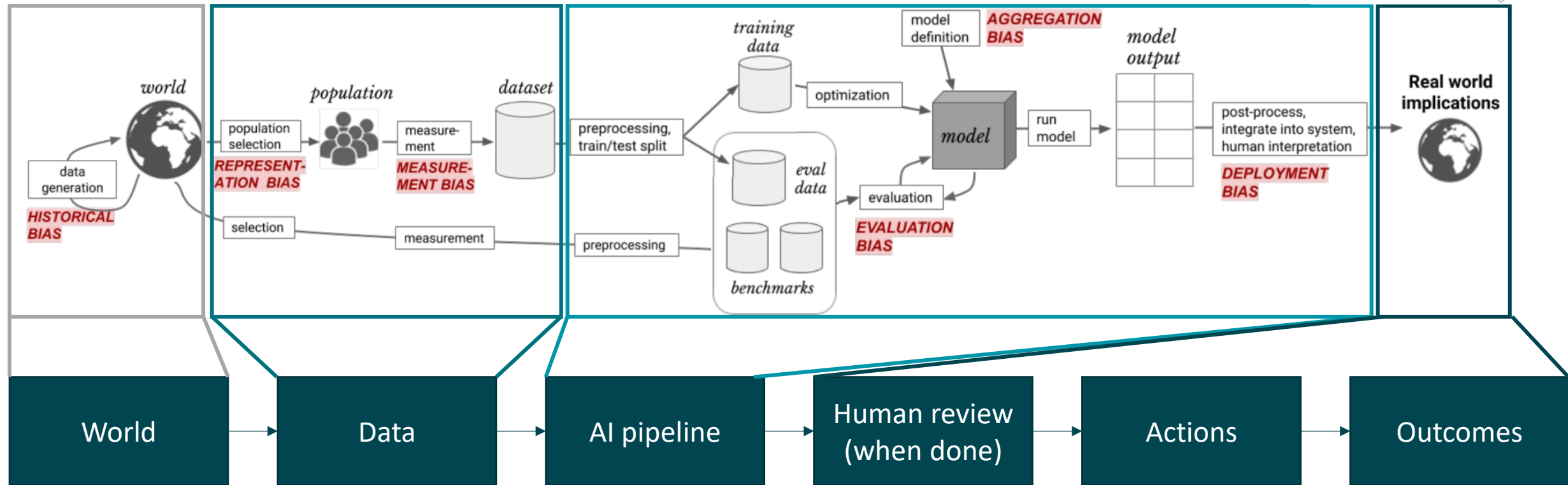
- Bias is about disparate errors against specific sub-groups.



Fairness

Bias in decision making

- Bias is about disparate errors against specific sub-groups.



Fairness

How to measure bias?

Sources: [1][2][3][4][5][6][7][8][9] view · talk · edit

		Predicted condition			
		Positive (PP)	Negative (PN)	Informedness, bookmaker informedness (BM) $= \text{TPR} + \text{TNR} - 1$	Prevalence threshold (PT) $= \frac{\sqrt{\text{TPR} \times \text{FPR}} - \text{FPR}}{\text{TPR} - \text{FPR}}$
Actual condition	Total population $= P + N$				
	Positive (P)	True positive (TP), hit	False negative (FN), type II error, miss, underestimation	True positive rate (TPR), recall, sensitivity (SEN), probability of detection, hit rate, power $= \frac{\text{TP}}{P} = 1 - \text{FNR}$	False negative rate (FNR), miss rate $= \frac{\text{FN}}{P} = 1 - \text{TPR}$
	Negative (N)	False positive (FP), type I error, false alarm, overestimation	True negative (TN), correct rejection	False positive rate (FPR), probability of false alarm, fall-out $= \frac{\text{FP}}{N} = 1 - \text{TNR}$	True negative rate (TNR), specificity (SPC), selectivity $= \frac{\text{TN}}{N} = 1 - \text{FPR}$
	Prevalence $= \frac{P}{P + N}$	Positive predictive value (PPV), precision $= \frac{\text{TP}}{\text{PP}} = 1 - \text{FDR}$	False omission rate (FOR) $= \frac{\text{FN}}{\text{PN}} = 1 - \text{NPV}$	Positive likelihood ratio (LR+) $= \frac{\text{TPR}}{\text{FPR}}$	Negative likelihood ratio (LR-) $= \frac{\text{FNR}}{\text{TNR}}$
	Accuracy (ACC) $= \frac{\text{TP} + \text{TN}}{P + N}$	False discovery rate (FDR) $= \frac{\text{FP}}{\text{PP}} = 1 - \text{PPV}$	Negative predictive value (NPV) $= \frac{\text{TN}}{\text{PN}}$ $= 1 - \text{FOR}$	Markedness (MK), deltaP (Δp) $= \text{PPV} + \text{NPV} - 1$	Diagnostic odds ratio (DOR) $= \frac{\text{LR+}}{\text{LR-}}$
	Balanced accuracy (BA) $= \frac{\text{TPR} + \text{TNR}}{2}$	F ₁ score $= \frac{2 \text{PPV} \times \text{TPR}}{\text{PPV} + \text{TPR}} = \frac{2 \text{TP}}{2 \text{TP} + \text{FP} + \text{FN}}$	Fowlkes–Mallows index (FM) $= \sqrt{\text{PPV} \times \text{TPR}}$	Matthews correlation coefficient (MCC) $= \frac{\sqrt{\text{TPR} \times \text{TNR} \times \text{PPV} \times \text{NPV}}}{-\sqrt{\text{FNR} \times \text{FPR} \times \text{FOR} \times \text{FDR}}}$	Threat score (TS), critical success index (CSI), Jaccard index $= \frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}}$

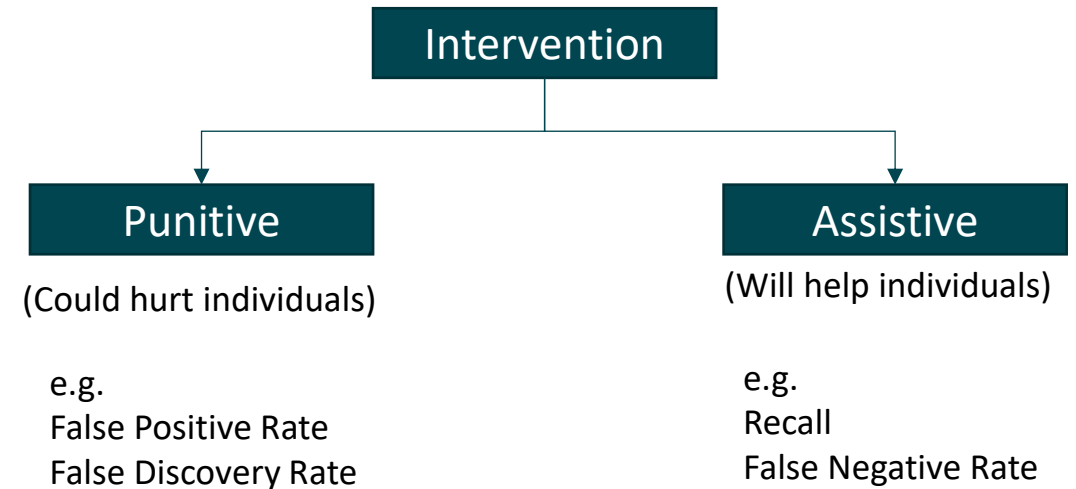
- Look for ration between groups

E.g.

$$\text{Disparity}_{\text{FNR}} = \frac{\text{FNR}_{\text{black}}}{\text{FNR}_{\text{white}}}$$

How to define the metric to use?

- If your intervention is **punitive in nature**, individuals may be harmed by intervening on them in error so you may care more about metrics that **focus on false positives**.
 - E.g. determining whom to deny a treatment
- If your intervention is **assistive in nature**, individuals may be harmed by failing to intervene on them when they have need, so you may care more about metrics that **focus on false negatives**.
 - E.g. determining who should receive an exercise voucher



Check the book for the full tree ☺ I am simplifying things here

Fairness

How to mitigate bias?

Pre-processing algorithms

- Relabeling
- Learned fair representation
- ...

In-processing algorithms

- Regularization
- Constrained optimization
- ...

Post-processing algorithms

- Threshold calibration
- ...



Audit a model's predictions

Requirements

Predictions or decisions
Labels
Sensitive attribute(s)

```
audit = Audit(df)
audit.summary_plot(metrics)
```

Correct a model's predictions

Requirements

Model object
Test set
Validation set
(recommended)

```
thresh = BalancedGroupThreshold(conf)
thresh.fit(X, preds_val, y, s)
thresh.transform(X, preds_test, s)
```

Training models w/ fairness considerations

Requirements

Dataset
Model configs (optional)

```
experiment = DefaultExperiment(df)
experiment.run()
```

<http://aequitas.dssg.io/>
<https://github.com/dssg/aequitas>

Privacy

Patient privacy should be a cornerstone principle while developing AI tools in medicine

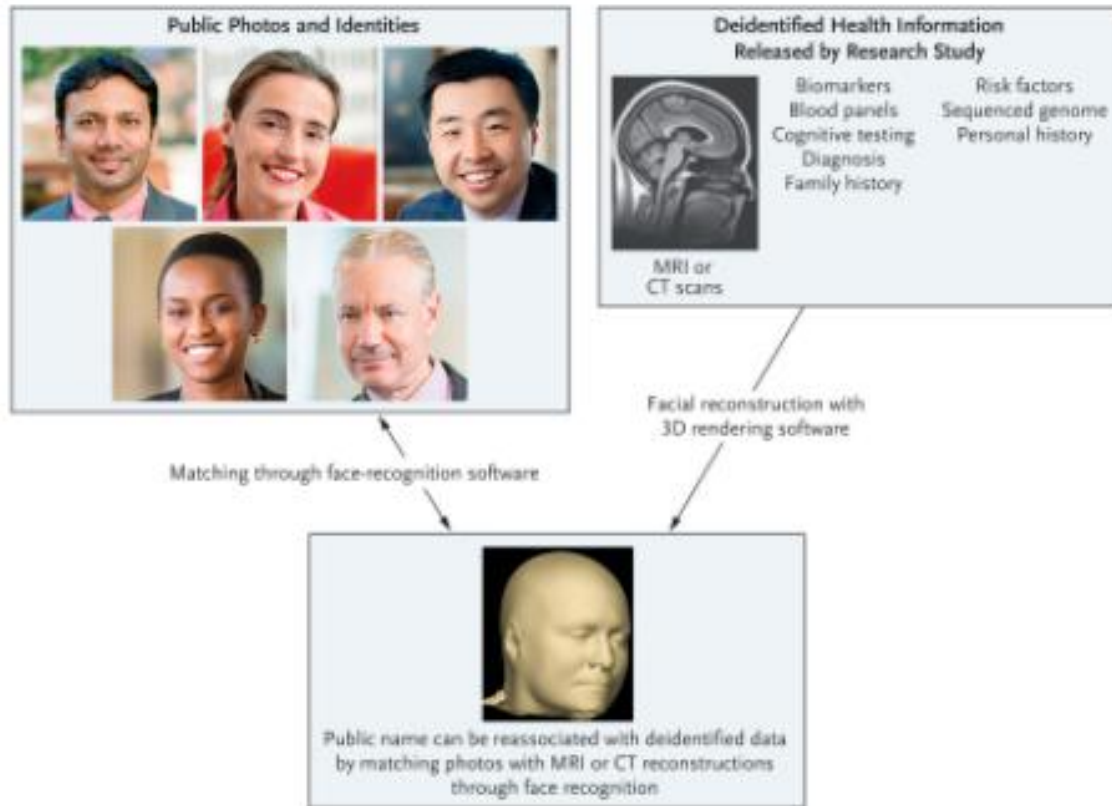


General
Data
Protection
Regulation



Privacy

Patient privacy should be a cornerstone principle while developing AI tools in medicine



Privacy

Patient privacy should be a cornerstone principle while developing AI tools in medicine

RESEARCH ARTICLE | NEUROSCIENCE



When makes you unique: Temporality of the human brain fingerprint

DIMITRI VAN DE VILLE , YOUNES FAROUJ , MARIA GIULIA PRETI , RAPHAËL LIÉGEAIS , AND ENRICO AMICO [Authors Info & Affiliations](#)

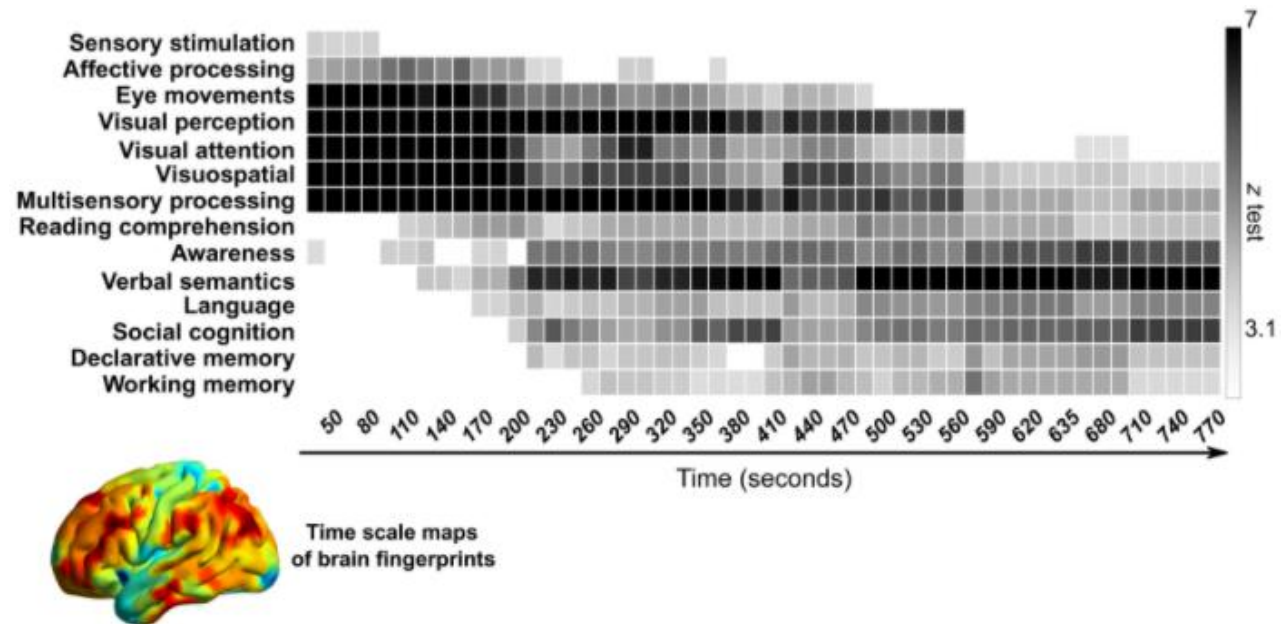
SCIENCE ADVANCES • 15 Oct 2021 • Vol 7, Issue 42 • DOI: 10.1126/sciadv.abj0751

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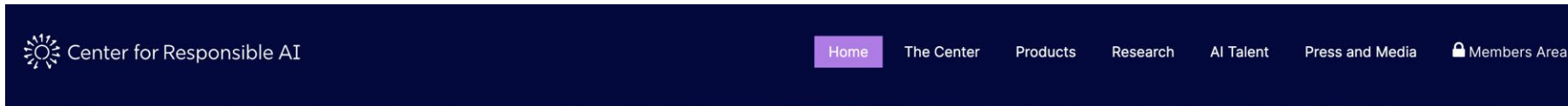


Abstract

The extraction of “fingerprints” from human brain connectivity data has become a new frontier in neuroscience. However, the time scales of human brain identifiability are still largely unexplored. We here investigate the dynamics of brain fingerprints along two complementary axes: (i) What is the optimal time scale at which brain fingerprints integrate information and (ii) when best identification happens. Using dynamic identifiability, we show that the best identification emerges at longer time scales; however, short transient “bursts of identifiability,” associated with neuronal activity, persist even when looking at shorter functional interactions. Furthermore, we report evidence that different parts of connectome finger-



Are you interested in the topic? In Portugal we are part of **Center for Responsible AI** aiming to set standards to use AI responsibly



10

Startups

8

Research Centers

2

Unicorns

1

Law Firm

5

Industry Leaders

21

Products

We believe in Fair, Explainable and Sustainable AI



Fair and transparent

We are committed to building AI products that help us build a more equal society.



Eco-friendly

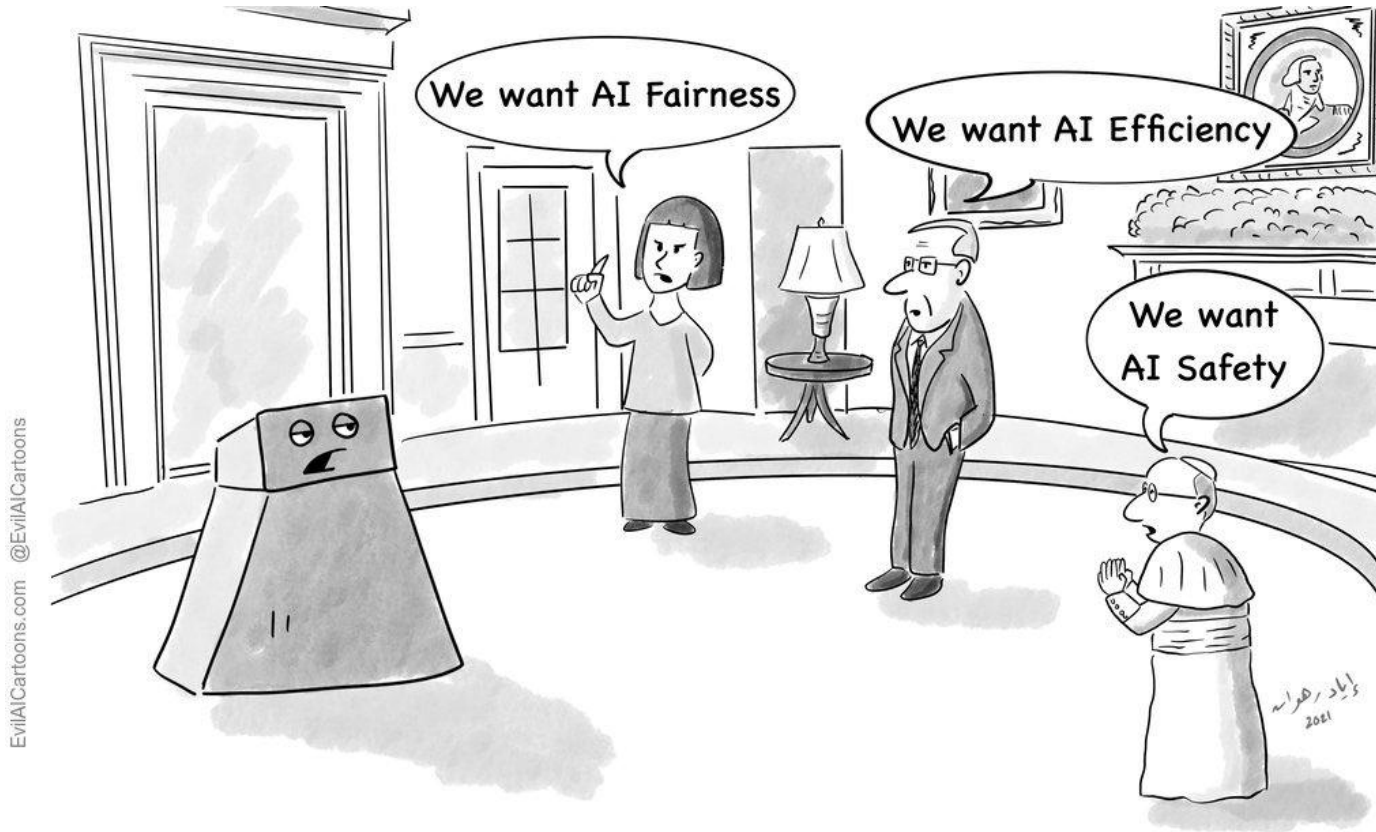
Developing AI algorithms that need less computing power, and are more sustainable.



Trustworthy

AI will not replace humans - it's a tool that can make us better. We are working to make AI more explainable and trustworthy.

Questions?



<< And I want infinite battery! Talk to me
when you've negotiated the tradeoffs! >>